

Workability and mechanical properties of alkali-activated materials: regulatory mechanisms dominated by composite alkali activator

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ABSTRACT

Alkali-activated materials (AAM) combine excellent mechanical performance and environmental benefits, making them promising alternatives to ordinary Portland cement. However, their application in high-performance concrete remains constrained by the unclear synergy between workability and mechanical properties. In this study, a composite activator system was developed using anhydrous Na_2SiO_3 as the base activator, supplemented with borax (NB), Na_2CO_3 (NC), and Na_2SO_4 (NS). The effects of Na_2SiO_3 modulus (M_s , 1.0–1.5), NB content (0–1.5 $\text{Na}_2\text{O}\%$), and NC/NS substitution levels (0–1.5 $\text{Na}_2\text{O}\%$) on the performance of AAM were systematically investigated through macroscopic performance tests and microstructural characterizations. Results indicate that NB effectively prolongs setting time and improves flowability of AAM, while low dosages (<1.0 $\text{Na}_2\text{O}\%$) also enhance mechanical strength. In Na_2SiO_3 systems, $M_s = 1.5$ provided a favorable balanced workability and strength, where as excessively low M_s induces flash setting. The incorporation of NC or NS generates transient phase precipitates during the early stages of reaction, thereby extending the setting time, but the associated reduction in free water negatively affects flowability. Low substitution ($\leq 0.375 \text{ Na}_2\text{O}\%$) of NC or NS promote a denser microstructure and higher compressive strength, while also contributing positively to sustainability. Furthermore, NC- Na_2SiO_3 composite systems impart a “high-strength-low-toughness” feature to AAM, making them suitable for developing Engineered geopolymer composites, whereas NS- Na_2SiO_3 systems are more applicable for enhancing the overall mechanical performance of concrete. This study reveals the synergistic regulation mechanisms of composite activators on AAM performance and provides both theoretical insights and practical references for the targeted optimization of high-performance concrete matrices.

1. Introduction

Alkali-activated materials (AAM) are widely regarded as promising alternatives to ordinary Portland cement (OPC), owing to their potential in enhancing energy efficiency, reducing carbon emissions in the construction industry, and enabling the high-value utilization of solid wastes [1,2]. AAM are generally produced through a series of chemical reactions between alkaline activators and aluminosilicate precursors [3,4]. The most commonly used precursors are supplementary cementitious materials (SCMs) that exhibit high pozzolanic activity when blended with OPC, such as ground granulated blast-furnace slag (GGBS), fly ash (FA), and metakaolin [5]. Under alkaline conditions, these precursors can dissolve and release reactive species such as Ca, Al, and Si at appropriate ratios, which subsequently re-polymerize in the solution to

form a dense gel matrix [6–8].

Fly ash (FA) and ground granulated blast-furnace slag (GGBS) are the most widely used precursors for alkali-activated materials (AAM). Their advantages are high availability, wide distribution, and stable chemical composition [9–13]. FA-based AAM usually show poor mechanical properties. High-temperature curing or high alkali content is often required to achieve acceptable strength [14,15]. These processes increase carbon emissions. For this reason, blended FA/GGBS systems are often adopted. They provide balanced mechanical and workability performance under ambient curing conditions [16–19]. Saha and Rajasekaran [20] showed that a higher slag content produces a denser matrix and higher compressive strength. Their blended AAM reached strengths up to 78.2 MPa. The performance of AAM depends not only on precursor ratios but also on the type and amount of activator. The most

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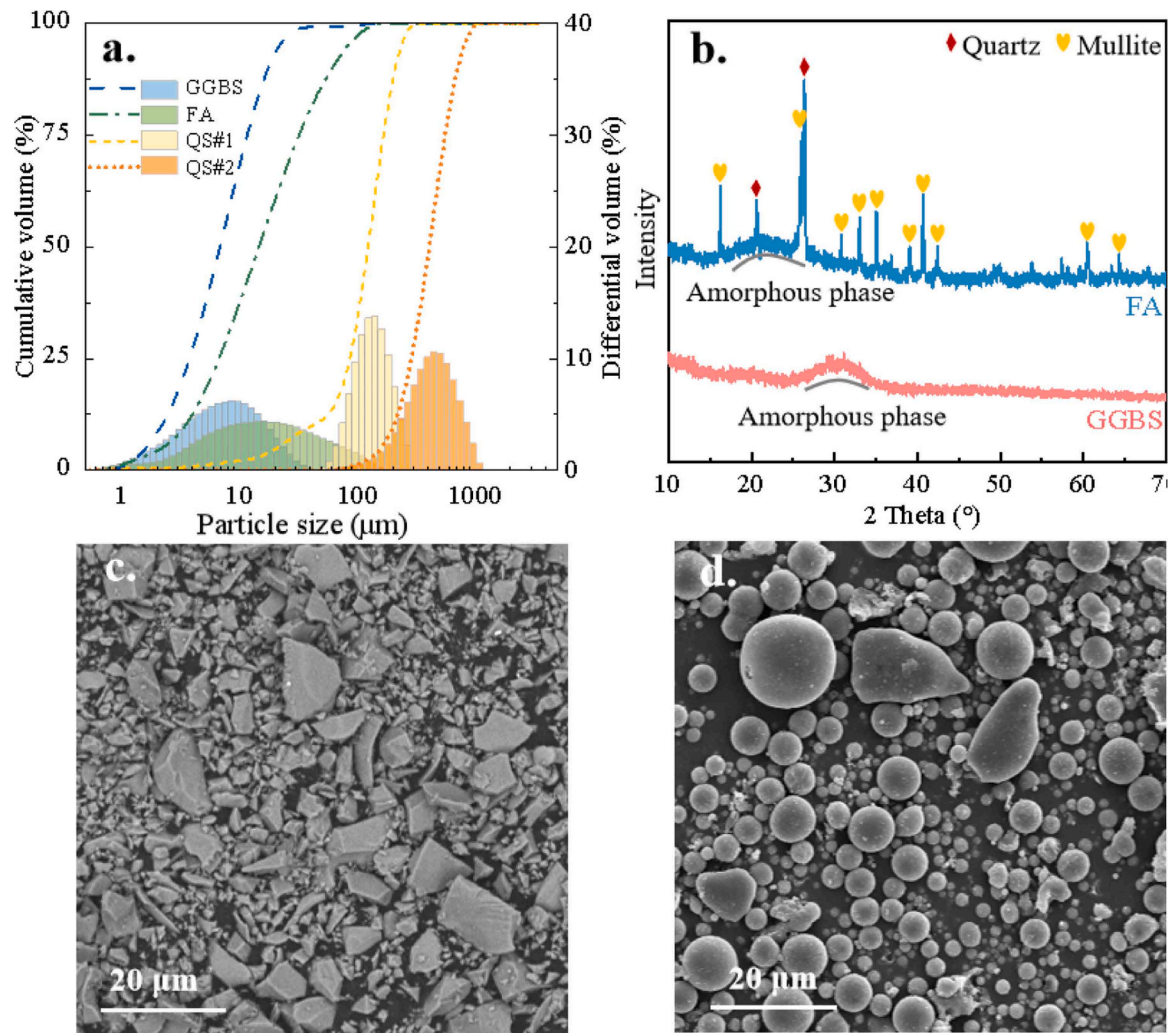


Fig. 1. Information on precursors and quartz sand materials. (a) Particle size distribution; (b) XRD patterns of the precursors; SEM patterns of (c) GGBS and (d) FA.

common activators are NaOH and Na_2SiO_3 [4,13,21]. In medium-calcium FA/slag systems, the combined use of NaOH and the cheaper Na_2SiO_3 improves strength and rheology. It does so without increasing carbon emissions or cost [16,22,23]. Phoo-ngernkham et al [16], compared NaOH, Na_2SiO_3 , and blended activators. At room temperature, the blended system gave the best strength development. The 28-day compressive strength of AASF reached 114.5 MPa. Lu et al [13], found that higher Na_2SiO_3 content increases the absolute value of the zeta potential, which enhances the initial fluidity of the paste. However, preparing blended NaOH - Na_2SiO_3 activators requires pre-mixing and cooling of the solutions, a time-consuming step that also complicates large-scale applications due to the handling of highly corrosive alkaline liquids. Solid activators, such as anhydrous Na_2SiO_3 , offer a practical alternative since they can be directly added with water during mixing, thereby mitigating many drawbacks of liquid activators [24]. Consequently, optimizing casting methods and developing “one-part” AAM may represent a more promising route, especially for in-situ construction.

The reaction mechanism of “one-part” AAM can still be described by the three-stage model of depolymerization, reorganization, and polymerization [25], but this process also involves an initial stage of activator dissolution. Strong alkaline activators such as NaOH and Na_2SiO_3 release considerable heat upon dissolution in water, which may cause flash setting when directly applied in AAM preparation [26]. Ma et al [26], attempted to partially replace anhydrous Na_2SiO_3 with Na_2CO_3 in AAM. Their results showed that sodium carbonate effectively extended

the initial setting time, maintained compressive strength above 50 MPa, and improved environmental sustainability. Consequently, milder solid activators such as NaAlO_2 , Na_2SO_4 , and Ca(OH)_2 have also been explored in AAM synthesis [24,27–29]. Systematic optimization of AAM performance is essential for the development of high-performance concrete. Taking engineered geopolymer composites as an example, to fully exploit their strain-hardening behavior and multiple micro-cracking, the matrix must simultaneously provide: (i) sufficient workability to ensure uniform fiber dispersion; (ii) a dense microstructure to enable effective fiber bridging; and (iii) relatively low fracture toughness to sustain steady-state cracking [30–33]. Therefore, performance-oriented design tailored to different application requirements has become a key challenge for developing high-performance concretes based on AAM as alternatives to Portland cement. However, systematic studies on the effects and mechanisms of different solid activators and their blends on the mechanical properties, workability, sustainability, and microstructure of AAM remain scarce. The lack of comprehensive cross-comparisons limits the broader application of these materials in engineering practice.

This study investigates the use of anhydrous Na_2SiO_3 , Na_2CO_3 , Na_2SO_4 , and $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ (borax) as blended solid activators for producing AAM with balanced performance. A combination of macroscopic tests and microstructural characterization was employed to systematically evaluate the effects and mechanisms of these activators. Macroscopically, the workability was determined by setting time and flowability (Sections 4.1 and 4.2), while mechanical performance was

Table 1

Chemical composition and physical parameters of slag and fly ash.

Materials	Oxide (wt%)								LOI (%)	Density (g/cm ³)
	CaO	SiO ₂	Al ₂ O ₃	SO ₃	Fe ₂ O ₃	MgO	Na ₂ O+K ₂ O	Others		
GGBS	41.97	28.49	15.53	3.1	0.47	8.19	0.92	1.83	0.84	3.1
FA	4.88	51.89	31.11	2.3	4.97	0.85	3.00	2.23	4.6	2.3

Note: LOI refers to loss on ignition.

evaluated through compressive strength and fracture property tests (Sections 4.3 and 4.4). The internal alkalinity of the AAM was also monitored (Section 4.5). Microstructural analysis involved XRD, SEM-EDS, and MIP to characterize the chemical composition, reaction degree, and pore structure of the hardened matrices (Section 4.6). Based on these findings, the underlying interaction mechanisms between the activators and the AAM matrix are discussed (Section 5), followed by a comprehensive assessment of mechanical performance and sustainability (Section 6).

2. Materials and mix proportions

2.1. Materials

The precursors used in this study were ground granulated blast-furnace slag (GGBS) and Class F low-calcium fly ash (FA), with median particle sizes (D_{50}) of 6.72 μm and 14.5 μm , respectively (Fig. 1a). The solid activators included anhydrous sodium silicate (with moduli of 1.0 and 2.0, corresponding to $\text{SiO}_2/\text{Na}_2\text{O}$ molar ratios of 1.0 and 2.0), sodium carbonate (NC), sodium sulfate (NS), and borax (NB). It should be noted that borax presents certain potential health hazards and requires careful handling during use. According to relevant chemical classification standards, borax may impair fertility or harm the unborn child, and may cause respiratory irritation. Therefore, strict personal protective measures—including the use of gloves, safety goggles, and dust masks—should be implemented throughout all preparation and handling procedures. A blend of quartz sands QS#1 and QS#2, with D_{50} values of 127 μm and 400 μm respectively, was used as fine aggregate. The rough surface of the QS, combined with a reduced liquid-to-solid ratio in the paste, provided the AAM with appropriate fluidity and plastic viscosity across different alkali-activators. This ensured uniform dispersion and complete dissolution of the solid activators. The chemical compositions of GGBS and FA were determined by X-ray fluorescence (XRF) analysis, and the results, together with density and loss on ignition, are presented in Table 1. X-ray diffraction (XRD) was employed to analyze the crystalline phases of the precursors, as shown in Fig. 1b. The FA exhibits significant amounts of quartz and mullite phases, along with a broad amorphous hump observed in the 15°–30° 2θ range. The crystalline and amorphous phase contents account for 57.2 % and 42.8 %, respectively.

In contrast, the GGBS shows no distinct crystalline diffraction peaks within the experimental resolution, with only a broad amorphous hump appearing between 24° and 36.5° 2θ . This indicates a predominantly amorphous structure, resulting in higher reactivity compared to FA.

2.2. Proportioning and preparation procedure

In this study, the combination and content of solid activators were used as the primary variables, while other mix parameters were fixed as follows: a water-to-binder ratio of 0.38; a GGBS-to-FA mass ratio of 1:1; total alkali content of 5 $\text{Na}_2\text{O}\%$; and a sand-to-binder ratio of 0.4, with the two quartz sands QS#1 and QS#2 blended at a mass ratio of 85:15. The activator system was designed by mixing anhydrous sodium silicate with moduli (M_s) of 1.0 and 2.0 to achieve different sodium silicate moduli (1.0, 1.25, and 1.5). Additionally, NB was added at different content (0.5, 1.0, and 1.5 $\text{Na}_2\text{O}\%$) based on equivalent Na_2O [17], and sodium carbonate (NC) and sodium sulfate (NS) were used to partially replace anhydrous sodium silicate at 0.375, 0.75, and 1.5 $\text{Na}_2\text{O}\%$. Herein, the equivalent Na_2O content is defined as the mass ratio of Na_2O to the precursors, which is calculated by converting the activators to an equimolar amount of Na_2O based on their sodium content. The detailed mix proportions of the AAM are summarized in Table 2. Both sand-containing and sand-free mixtures are classified as AAM and are referred to as “mortar” and “paste,” respectively.

The specific preparation and curing procedures for AAM mortar were as follows. First, the precursors, aggregates, and a solid activator were placed in a planetary mixer and dry-mixed at 75 r/min for 3 min to ensure uniformity. Then, tap water was added, and mixing continued for another 3 min to obtain a homogeneous paste. The fresh paste was cast into molds and compacted on a vibration table. After the initial setting, the specimens were covered with plastic film and cured under laboratory conditions for 24 h. After demolding, the specimens were cured under standard conditions until the testing ages of 3, 7, and 28 days for mechanical property evaluation.

The prepared specimens consisted of 50 mm $\times$ 50 mm $\times$ 50 mm cubes for compressive strength tests and 40 mm $\times$ 40 mm $\times$ 160 mm prisms for fracture tests. In addition, sand-free pastes were prepared following the same procedure for flowability, setting time, pH, and XRD

Table 2AAM mix proportions (kg/m³).

Mix IDs	GGBS	FA	Na ₂ O-SiO ₂	Na ₂ O-2SiO ₂	Na ₂ CO ₃	Na ₂ SO ₄	Borax	Water	QS#1	QS#2
M1.5-B0	515.0	515.0	52.33	78.91	/	/	0.00	441.3	350.2	61.80
M1.5-B1			52.33	78.91	/	/	31.73	432.7		
M1.5-B2			52.33	78.91	/	/	63.46	424.1		
M1.5-B3			52.33	78.91	/	/	95.19	415.6		
M1.0-B2			104.7	0.00	/	/	63.46	414.0		
M1.25-B2			78.50	39.46	/	/	63.46	419.1		
M1.5-B2-NC1			48.41	72.99	6.600	/	63.46	422.9		
M1.5-B2-NC2			44.48	67.07	13.21	/	63.46	421.7		
M1.5-B2-NC3			36.63	55.24	26.41	/	63.46	419.2		
M1.5-B2-NS1			48.41	72.99	/	8.850	63.46	423.7		
M1.5-B2-NS2			44.48	67.07	/	17.69	63.46	423.4		
M1.5-B2-NS3			36.63	55.24	/	35.39	63.46	422.6		

Note: Mx-By denotes an AAM with a sodium silicate modulus of x and a borax (NB) content of y, where x = 1.0, 1.25, and 1.50 correspond to different sodium silicate moduli, and y = 0, 1, 2, and 3 represent borax contents of 0, 0.5, 1.0, and 1.5 $\text{Na}_2\text{O}\%$, respectively. Mx-By-NC/NSz indicates an AAM containing sodium carbonate (NC) or sodium sulfate (NS) at z, where z = 1, 2, and 3 correspond to NC/NS contents of 0.375, 0.75, and 1.5 $\text{Na}_2\text{O}\%$, respectively.

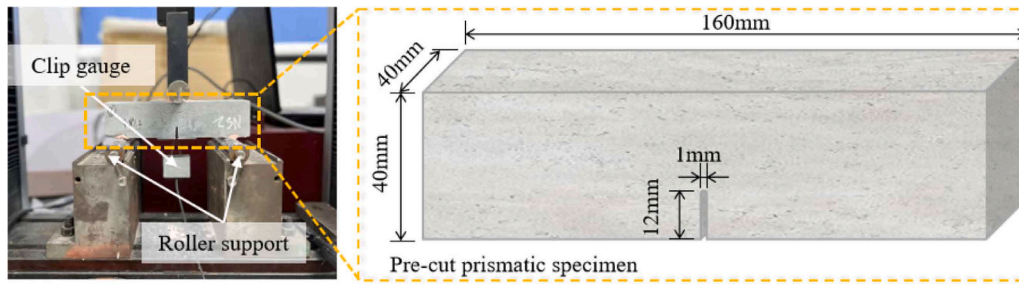


Fig. 2. Fracture test setup.

analyses.

3. Experimental procedure

3.1. Workability test

The setting time of the sand-free pastes was determined using a Vicat apparatus in accordance with the Chinese standard *Methods of Test for Water Requirement, Setting Time and Soundness of Cement* (GB/T 1346–2011 [34]). The tests were conducted in an air-conditioned room at $20 \pm 2^\circ\text{C}$, and the penetration depth of the needle was recorded until final setting. Initial setting was defined when the needle penetration reached $4 \text{ mm} \pm 1 \text{ mm}$ from the base plate, and final setting was defined when the penetration depth was less than 0.5 mm .

The flowability of the fresh pastes was also measured according to *Test Methods for Homogeneity of Concrete Admixtures* (GB/T 8077–2012 [35]). A truncated cone mold with a top diameter of 36 mm, a bottom diameter of 60 mm, and a height of 60 mm was used. The flow spread was determined by taking the average of the vertical diameters after the paste ceased to flow.

3.2. Mechanical properties test

Compressive strength tests of mortar cubes at different standard curing ages (3, 7, and 28 days) were conducted in accordance with ASTM C109/C109M-2020a [36]. The tests were performed using a hydraulic testing machine (C088–01, Matest, Italy) at a loading rate of 1.0 kN/s . In addition, 28-day prisms were used for three-point bending fracture tests. Prior to testing, a single-edge notch with a width of 1 mm and a height of 12 mm was pre-cut at the midspan of each prism. Aluminum strips with a thickness of 0.3 mm were attached on both sides of the notch to fix the clip-on extensometer for measuring crack mouth opening displacement (CMOD). The span was set to 120 mm, and roller supports were used to minimize the influence of lateral friction on the results. Loading was applied at a rate of 0.02 mm/min . The experimental setup is shown in Fig. 2.

3.3 pH tests

The pH of the activator solutions, fresh pastes, and hardened pastes at different curing ages (7 and 28 days) was measured. Activator solutions were prepared by dissolving the solid activators from the corresponding mix proportions in water. For both fresh paste and hardened bulk samples, the pH value was determined using ex situ measurement methods based on previously established research [37–40]. For fresh paste pH measurement, powder samples were prepared by mixing the

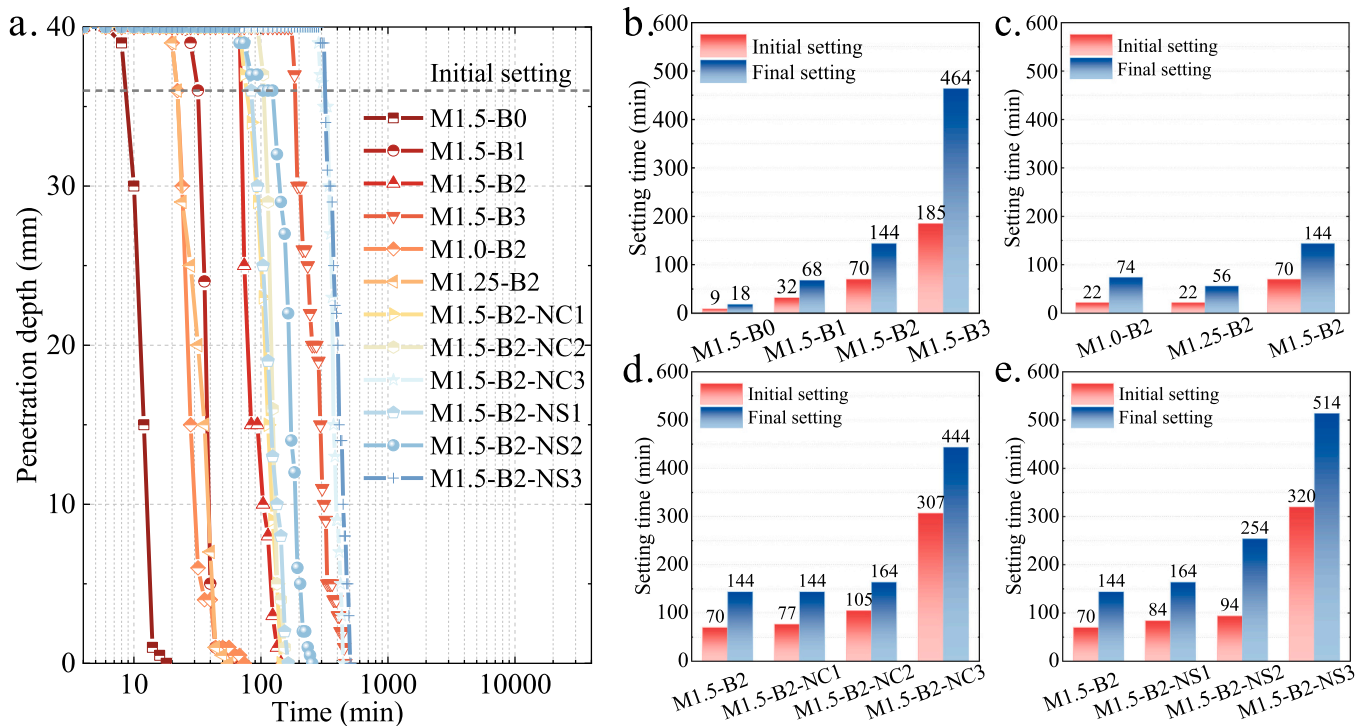


Fig. 3. Effect of different activator systems on setting time. (a) Penetration depth vs. time curve; (b-e) Individual effects on $ST_{initial}$ and ST_{final} : (b) NB content, (c) M_s , (d) NC content, and (e) NS content.

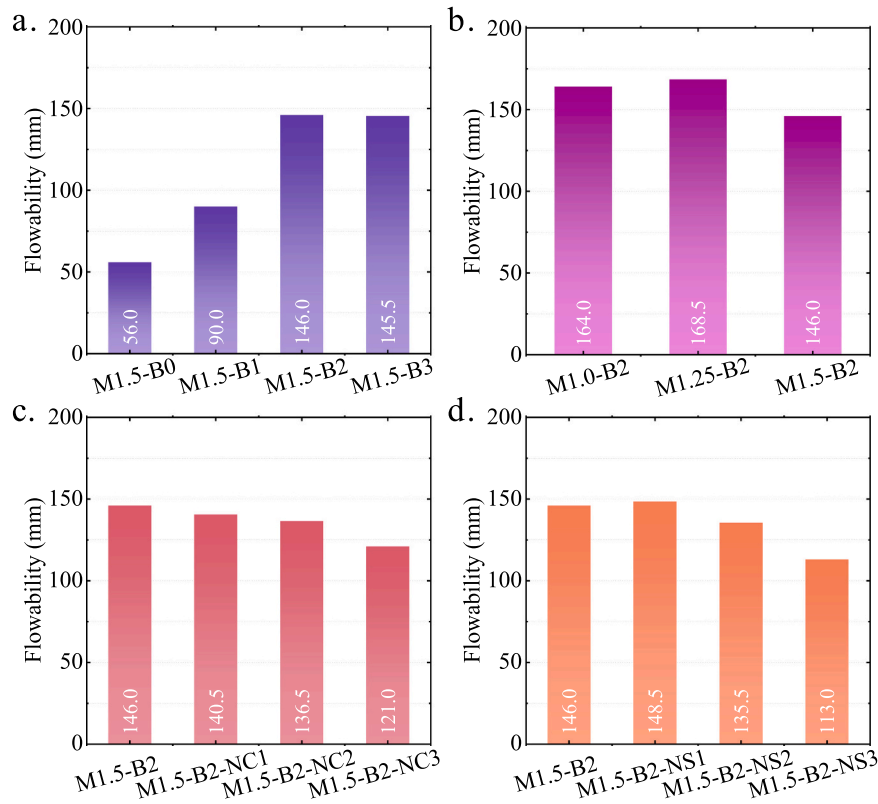


Fig. 4. The effect of various activator system parameters on the paste flowability: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

fresh paste with isopropanol at a 1:50 mass ratio in a beaker to form a suspension, thereby stopping hydration. After 24 h of sedimentation, the lower layer was collected. Most isopropanol was removed under vacuum, and the remaining material was dried in an oven at 50 °C for 10 h. The dried sample was passed through a 200-mesh sieve to obtain uniform powder. Hardened paste samples were prepared similarly, except that the specimens were first cured for 7 and 28 days under standard conditions, broken into pieces, treated with isopropanol to stop hydration, followed by vacuum removal and drying, and finally ground through a 200-mesh sieve.

Powder samples were then mixed with deionized water at a 1:1 mass ratio to form a solution, which was filtered through Whatman qualitative filter paper with an 11 μm pore size to obtain a clear filtrate. pH measurements of the activator solutions and the filtrates were conducted directly using a pH meter in an air-conditioned room at 20 °C. It is noteworthy that the pH value was recorded when the rate of change of the pH meter reading fell below 0.01 unit per minute.

3.3. Microstructural characterization

Hardened mortar specimens at 28 days were broken and immersed in isopropanol. After vacuum drying, the samples were divided into two groups for microstructural characterization:

- (1) Samples with natural fracture surfaces were gold-coated and directly observed by scanning electron microscopy (SEM) to examine the microstructure and the formation of hydration products;
- (2) Mortar fragments were embedded in resin molds, cured, and polished with SiC sandpaper to obtain a smooth surface. During polishing, isopropanol was applied repeatedly as a lubricant to minimize additional hydration caused by overheating. The polished samples were gold-coated and observed under SEM, and energy-dispersive spectroscopy (EDS) was employed for area and

point scans to obtain elemental distributions, including Ca, Si, and Al. For each group, six high-resolution images were captured in backscattered electron mode to evaluate the degree of reaction of the cementitious material.

To eliminate interference from quartz aggregates, powder samples prepared from sand-free hardened pastes were used for X-ray diffraction (XRD) analysis. The 28-day hardened blocks were ground in an agate mortar, passed through a 200-mesh sieve, and dried prior to XRD testing. Measurements were performed at a scanning rate of 2°/min over a 10°–70° 2 θ range. Additionally, hardened pastes were cut into small cubes with side lengths ≤ 10 mm for mercury intrusion porosimetry (MIP) to analyze pore structure.

4. Test results

4.1. Setting time

Fig. 3 illustrates the effects of different activator systems on the setting time. Fig. 3a shows the penetration depth of the Vicat needle as a function of time. It can be observed that the penetration depth decreased over time, and the trend varies significantly among different activator combinations, indicating the rationality of the chosen experimental variables. Notably, the M1.5-B0 sample exhibited the earliest setting, suggesting a pronounced retarding effect of NB. Based on Fig. 3a, the initial and final setting times ($ST_{initial}$ and ST_{final}) were determined for each system and are presented in Fig. 3b–e.

Fig. 3b shows the effects of different NB contents on the setting time of alkali-activated materials. The results indicate that both initial and final setting times increased with increasing NB content. Without NB, the paste exhibited flash setting, with an $ST_{initial}$ of only 9 min. With 1.0 Na₂O% NB, the $ST_{initial}$ was extended to 70 min, exceeding the standard requirement of 45 min [41], and the ST_{final} increased to 144 min. This can be mainly attributed to: (1) NB reacting with Ca²⁺ on the slag

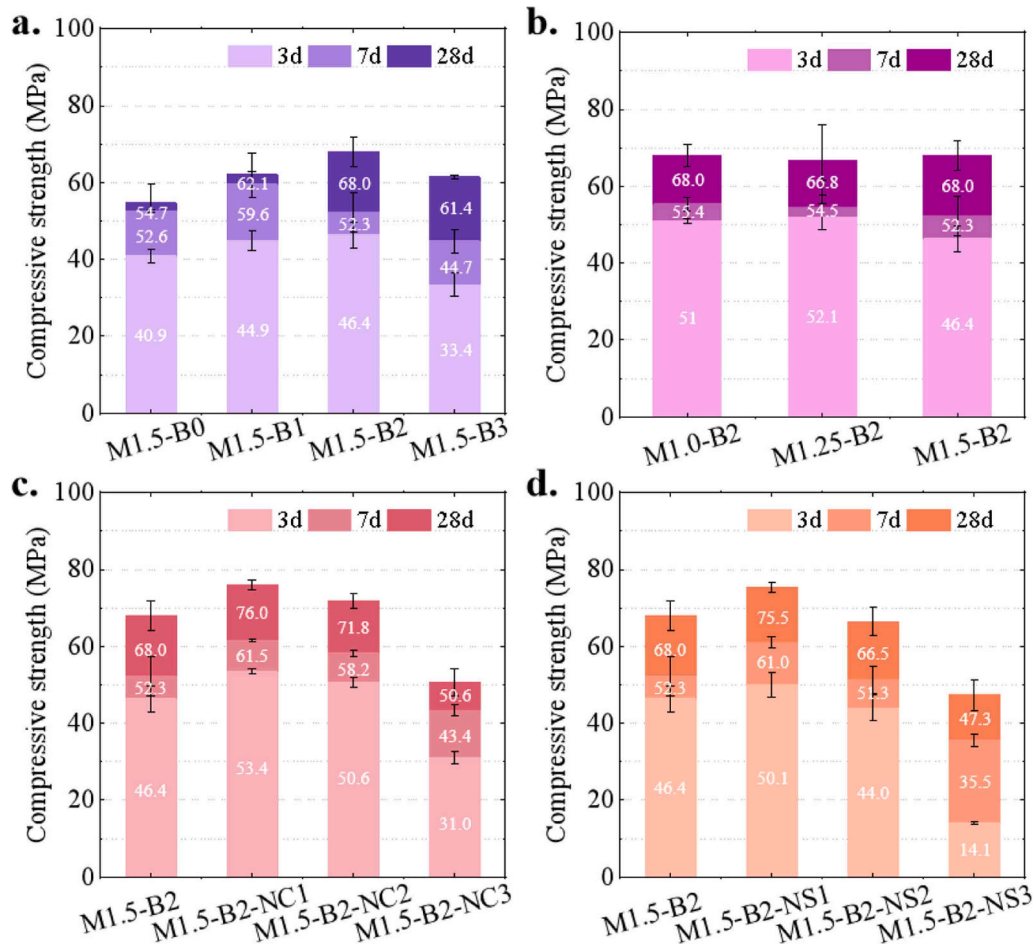


Fig. 5. The effect of various activator system parameters on compressive strength of AAM: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

surface to form calcium borate, which inhibits the release of Ca^{2+} into the pore solution [42,43]. and (2) borate impeding the formation of Si-O-Al bonds, thereby slowing down the early-stage reaction [44]. Fig. 3c illustrates the setting times of AAM at different sodium silicate moduli (M_s). Lower M_s values corresponded to shorter setting times. For M1.0-B2 and M1.25-B2, the ST_{initial} was 22 min, while the ST_{final} were 74 min and 56 min, respectively. Figs. 3d and 3e present the effects of different amounts of NC and NS on the setting time of AAM. Both NC and NS delayed the setting process, with NS showing a more pronounced retarding effect. This is primarily attributed to the formation of intermediate precipitates from CO_3^{2-} and SO_4^{2-} reacting with Ca^{2+} , and it hinders the dissolution and polycondensation of silicoaluminates [45].

4.2. Flowability

Fig. 4 illustrates the effects of different activator systems on the fluidity of the pastes. Fig. 4a presents the influence of NB content on the flow diameter. The results show that the fluidity of the paste increased significantly with increasing NB content. At a content of 1.0 $\text{Na}_2\text{O}\%$, the flow diameter of M1.5-B2 was 160.7 % higher than that of M1.5-B0. This improvement is mainly attributed to the strong retarding effect of NB. In the absence of NB, M1.5-B0 underwent flash setting (see Fig. 4b), and the premature formation of hydration products severely reduced its fluidity. When the NB content exceeded 1.0 $\text{Na}_2\text{O}\%$, the flow diameter no longer increased. This phenomenon is related to the calcium borate protective layer formed on the precursor surface, which imparts a negative zeta potential [42], enhances particle dispersion, and thereby improves the workability of AAM.

Fig. 4b shows the effect of M_s on paste fluidity. It can be observed

that the flow diameter decreased with increasing M_s , which is consistent with the findings of Zhang et al. [46]. The reduction in fluidity at higher M_s values can be attributed to the growth of cluster structures in the fresh paste, impairing the dispersion of particles/aggregates and consequently decreases paste fluidity [47].

Figs. 6c and 6d illustrate the effects of different NC and NS contents on the fluidity of AAM pastes. The results indicate that the incorporation of both NC and NS reduced the flow diameter. When NC was used as a supplementary activator to partially replace sodium silicate, the paste exhibited a noticeable reduction in fluidity, which is consistent with the findings reported by Lu et al. [13]. This phenomenon can be attributed to the decreased solubility of NC under highly alkaline conditions due to the common ion effect, leading to the formation of crystalline precipitates such as $\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$ or the generation of transitional phases such as $\text{Na}_2\text{Ca}(\text{CO}_3)_2 \cdot 5 \text{H}_2\text{O}$ through interaction with dissolved Ca^{2+} , thereby reducing the amount of free water in the system [3,48]. In addition, when NS was incorporated, competitive adsorption of Ca^{2+} by SiO_3^{2-} and SO_4^{2-} occurred within the paste. SO_4^{2-} preferentially reacted with Ca^{2+} to form early gypsum ($\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$) precipitates, the $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$ can similarly consumed free water and consequently decreased the fluidity of the AAM paste [45].

4.3. Compressive strength

Fig. 5 illustrates the evolution of compressive strength (σ_c) of alkali-activated mortars with different activator systems. Overall, the compressive strength increased with curing age for all systems. Fig. 5a shows the effect of NB content. When the NB content was 1.0 $\text{Na}_2\text{O}\%$, the 28-day compressive strength reached the maximum value of 68.0 MPa,

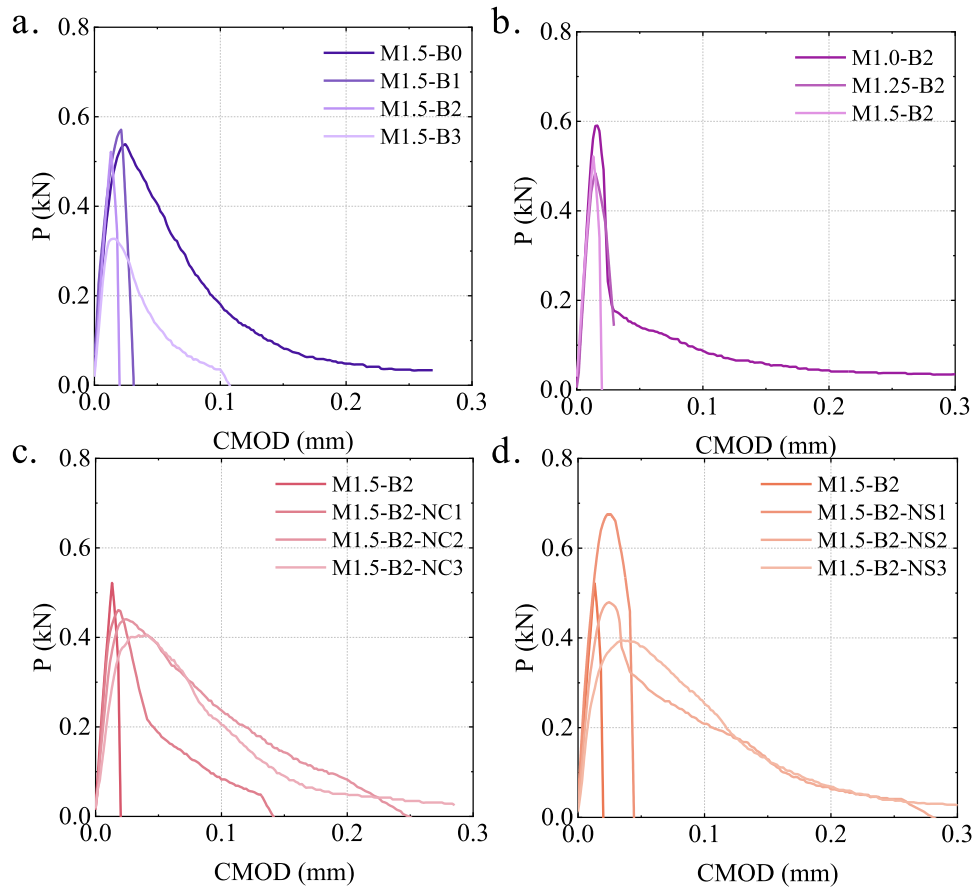


Fig. 6. Three-point bending P-CMOD curves of AAM with various activator system parameters: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

which was 29.3 % higher than that of the control group (M1.5-B0). This indicates that NB can partially substitute for Al to participate in the polymerization and hydration reactions, and a moderate content is beneficial for strength development [49]. However, when the NB content increased to 1.5 Na₂O%, the 28-day strength was slightly lower than that of M1.5-B2, and the ratio of strength gain was significantly reduced. This is likely due to excessive NB lowering the alkalinity of the pore solution, thereby suppressing the formation of reaction products.

As shown in Fig. 5b, when the M_s ranged from 1.0 to 1.5, no obvious difference in 28-day compressive strength was observed. Notably, mortars with lower M_s exhibited a relative advantage in early-age strength development. Ravikumar et al [50]. reported that, at an alkali dosage of 5.0 Na₂O%, the 28-day compressive strength increased with increasing M_s . The discrepancy may be attributed to the insufficient dissolution of solid activators at ambient temperature, which affected the effective alkalinity.

Figs. 7c and 7d present the effects of NC and NS incorporation on compressive strength. At a content of 0.375 Na₂O%, the 28-day compressive strengths of mortars with NC and NS increased by 11.8 % and 11.0 %, respectively, compared to the control group (M1.5-B2). The improvement can be ascribed to the formation of calcite in the presence of NC and the generation of ettringite in the case of NS, both of them could enhance mechanical performance [51,52]. However, further increasing the NC/NS content led to a gradual reduction in strength, which was mainly due to the decreased alkalinity of the activator system. The lower alkalinity would hinder precursor dissolution and reaction, and ultimately resulting in strength loss.

4.4. Fracture performance

Fig. 6 illustrates the P-CMOD curves of alkali-activated mortars

under three-point bending with different activator systems. As shown in Fig. 6a, the incorporation of NB increased the brittleness of the mortar, leading to immediate brittle fracture after reaching the peak load. Fig. 6b demonstrates that mortars with $M_s = 1.0$ exhibited pronounced post-peak ductile behavior, whereas increasing M_s adversely affected post-peak ductility, resulting in more brittle fracture behavior. Fig. 6c shows that NC improved the post-peak ductility of mortars, with ductility consistently increasing as the NC content increased. Fig. 6d reveals that NS exhibited a similar effect to NC. However, when the NS content was 0.375 Na₂O%, the peak load was substantially enhanced, the fracture behavior still remained brittle though.

Fracture energy (G_f) and fracture toughness (K_{IC}) are key parameters for characterizing the fracture behavior of materials. G_f represents the energy consumed during the fracture process, defined as the total energy required to create a unit area of crack surface. It reflects the total energy absorbed by the material from crack initiation to complete failure, serving as a critical indicator for evaluating toughness and energy dissipation capacity. K_{IC} denotes the threshold of a material's resistance to unstable crack propagation, corresponding to the critical stress intensity factor in linear elastic fracture mechanics. When the stress intensity reaches this value, unstable crack growth occurs, making this parameter essential for assessing the load-bearing safety of materials containing defects. In material design, higher fracture energy helps suppress crack propagation in concrete and enhances its ductility, while appropriately reduced fracture toughness in Engineered cementitious composites/Engineered geopolymer composites matrix materials facilitates multiple cracking behavior and improves deformation capacity. The fracture energy G_F of alkali-activated mortar was calculated using the formula recommended by RILEM:

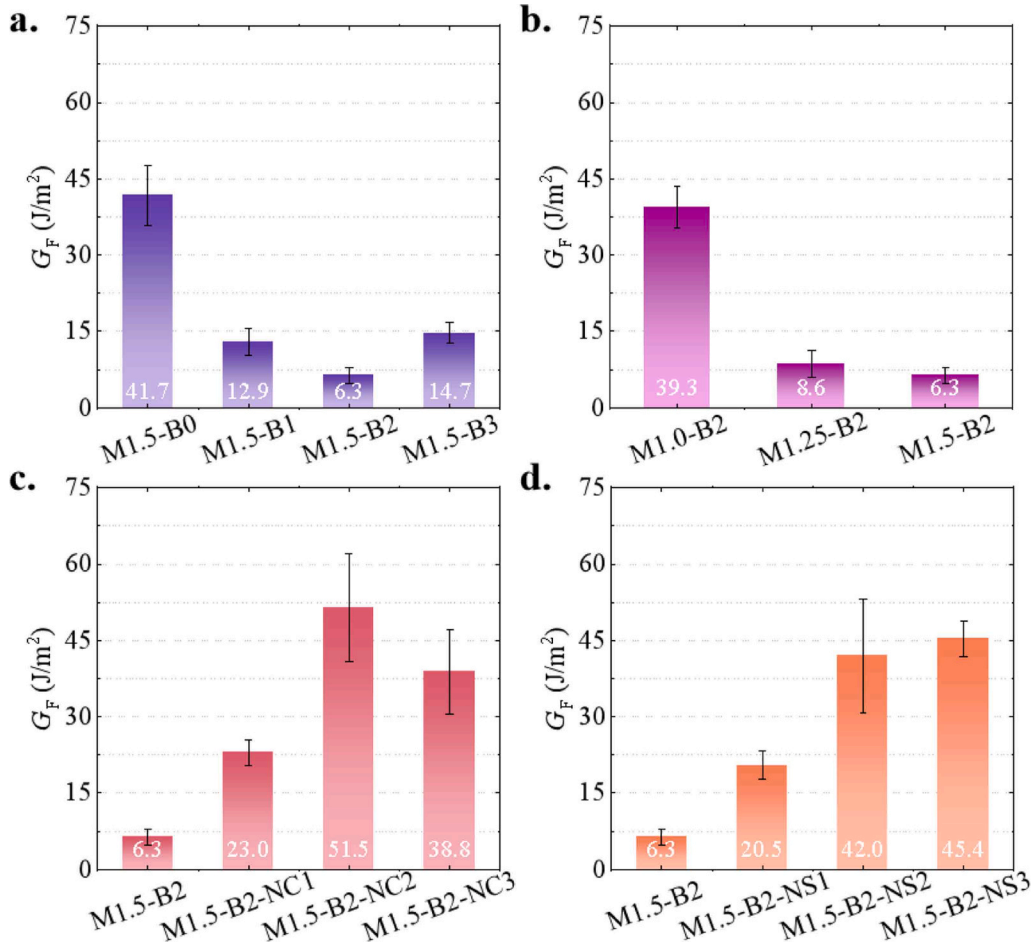


Fig. 7. Fracture energy of AAM with various activator system parameters: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

$$\text{Here, } w_0 \text{ is the area under the } G_F = \frac{w_0 + mg\delta_0}{A_{lig}} = \frac{\int_0^{\delta_0} P(\delta) d\delta + mg\delta_0}{W(B-a)} \quad (1)$$

load-displacement curve (J), m is the specimen mass (kg), g is the gravitational acceleration (m/s^2), δ_0 is the mid-span deflection at peak load (μm), A_{lig} is the fracture surface area of the specimen (mm^2), W and B are the thickness and height of the notched specimen (mm), and a is the pre-crack depth (mm). In this study, the area under the Load-CMOD curve was used to calculate w_0 , and the peak crack mouth opening displacement ($CMOD_0$) was used instead of δ_0 to calculate the work of gravity. Under three-point bending, the mid-span deflection of a notched beam is linearly correlated with $CMOD_0$, as confirmed in previous studies [53], providing a basis for calculating the fracture energy here. It should be noted that the fracture energy calculated using $CMOD_0$ is generally lower than that calculated with δ_0 , but it remains suitable for comparing fracture energies between different mortars. The calculated fracture energies of AAM specimens are presented in Fig. 7.

The incorporation of NB and the increase of M_s both adversely affected the ductility of the mortars, as reflected by the reduction in fracture energy, as shown in Figs. 9a and 9b. Regarding the effect of M_s , similar trends were reported by Zhang et al [54], where the fracture energy reached a maximum at $M_s = 1.0$ and decreased with increasing M_s . Figs. 9c and 9d illustrate the enhancing effect of NC and NS on fracture energy. Previous studies by Kendall et al [55], indicated a significant correlation between fracture energy and porosity. The addition of NC and NS reduced the mortar porosity (Section 4.7.4), consistent with the observed increase in fracture energy.

The fracture toughness of the alkali-activated mortar was calculated according to the formula provided in ASTM E399-19 [56]:

$$K_m = \frac{P_s S}{WB^{3/2}} \cdot 3 \sqrt{\frac{a}{B}} \frac{1.99 - \left(\frac{a}{B}\right) \left(1 - \frac{a}{B}\right) \left[2.15 - 3.93 \frac{a}{B} + 2.7 \left(\frac{a}{B}\right)^2\right]}{2 \left(1 + 2 \frac{a}{B}\right) \left(1 - \frac{a}{B}\right)^{3/2}} \quad (2)$$

Here, P_s denotes the peak load in the fracture test (N), and S is the span of the specimen (mm). The results are shown in Fig. 8.

The fracture toughness reached its maximum at a NB content of 0.5 $Na_2O\%$, representing a 21.0 % increase compared with M1.5-B0. Further increasing the NB content led to a reduction in fracture toughness, as shown in Fig. 8a. In particular, M1.5-B3 exhibited a 40.6 % decrease relative to the control, this can be attributed to the reduced reaction degree caused by the lower pore solution pH of the alkali-activated mortar. As illustrated in Fig. 8b, the fracture toughness of mortars was highest at $M_s = 1.5$, which differs from the results reported by Zhang et al [54], for alkali-activated materials. This discrepancy may be attributed to the dominant role of binder-quartz sand interactions in governing the fracture toughness of mortars within the M_s range of 1.0–1.5. The effect of partially replacing sodium silicate with NC on fracture toughness differed markedly from its influence on compressive strength, as fracture toughness decreased consistently with increasing NC content (Fig. 8c). In contrast, the incorporation of NS produced a non-monotonic trend, with fracture toughness first increasing and then decreasing (Fig. 8d). The maximum value of $0.389 MPa \cdot m^{1/2}$ was achieved at an NS content of 0.375 $Na_2O\%$.

4.5 pH tests

pH tests were conducted on alkali-activator solutions, fresh pastes, and hardened AAM specimens under different activator systems, as shown in Fig. 9. The results indicate that blended activators containing

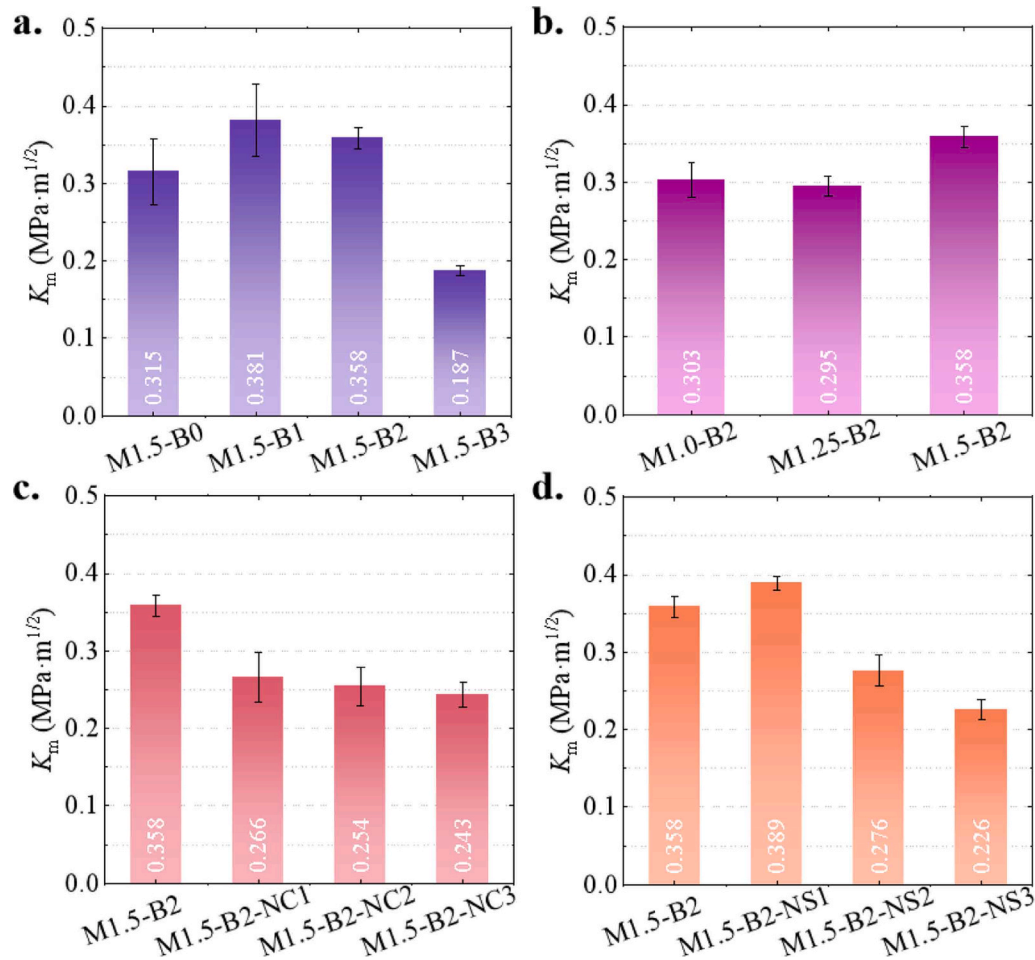


Fig. 8. Fracture toughness of AAM with various activator system parameters: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

NB, NC, or NS exhibited lower pH values compared to pure sodium silicate, and the pH further decreased with increasing content of these supplementary activators. In addition, Fig. 9b shows that higher M_s values also reduced the pH of the activator solution.

As illustrated in Fig. 9a, when the NB content was below 1.0 $\text{Na}_2\text{O}\%$, the pH of fresh pastes was lower than that of the corresponding activator solution. This is attributed to the insufficient coverage of precursors by Ca-borate phases, enabling greater participation of sodium silicate in the reaction and consequently lowering the system pH. This observation is consistent with the shorter setting times recorded for these mixtures. Similarly, in Fig. 9b, activator solutions with low M_s (<1.5) exhibited higher pH values than their corresponding pastes. This may be due to the higher solubility of low- M_s sodium silicate compared with the relatively lower solubility of NB under the common-ion effect. In fresh pastes, insufficient Ca-borate coverage facilitated direct reactions between sodium silicate and the precursors, lowering the paste pH and leading to shorter setting times. In contrast, at $M_s > 1.25$, the consumption of NB enhanced sodium silicate solubility, resulting in fresh pastes with higher pH than the activator solution. These results suggest that the retardation effect of NB is less pronounced in low- M_s (i.e., high-alkalinity) systems. For systems incorporating NC and NS, the fresh pastes consistently exhibited higher pH values than the corresponding activator solutions (Figs. 11c and 11d). This can be attributed to the preferential reactions of CO_3^{2-} and SO_4^{2-} with Ca^{2+} , which compete with SiO_3^{2-} for Ca^{2+} binding. Furthermore, Fig. 9 shows that the pH of hardened AAM was lower than that of fresh pastes and decreased progressively with curing age, corresponding to the development of mechanical properties.

4.5. Microstructural characterization

4.5.1. Microstructural analysis based on SEM

Fig. 10 presents the microstructural morphologies of AAM under different activator systems. As shown in Fig. 10a, the microstructure of the AAM activated solely with sodium silicate exhibited a rough and uneven gel structure. In contrast, the fracture surface of M1.5-B2 with NB addition (Fig. 10b) appeared relatively smooth and flat, indicating a transition of the failure mode from brittle fracture to cleavage fracture upon NB incorporation [57]. This transition lowers the energy absorption capacity during fracture, accounting for the reduced fracture energy with increasing NB content.

Fig. 10c shows the microstructure of M1.0-B2 with $M_s = 1.0$, where abundant traces of C-S-H gel formation can be observed. Compared with M1.5-B2, the gel structure of M1.0-B2 appeared denser and smoother, but larger microcracks were present. At the macroscopic level, this corresponds to an increase in fracture energy but a reduction in fracture toughness.

With the incorporation of NC, the microstructure of the AAM became denser, the porosity decreased (as confirmed in Section 4.6.4), and microcracks were refined (Figs. 12d and 12e). The densified microstructure enhanced both compressive strength and fracture energy. However, the increased number of microcracks within the gel structure promoted stress concentration under tensile loading, leading to reduced fracture toughness. In particular, at an NC content of 1.5 $\text{Na}_2\text{O}\%$ (M1.5-B2-NC3), a large amount of CaCO_3 crystals was observed. The formation of CaCO_3 , a relatively low-density phase, appears to be the main cause of the extensive microcracking.

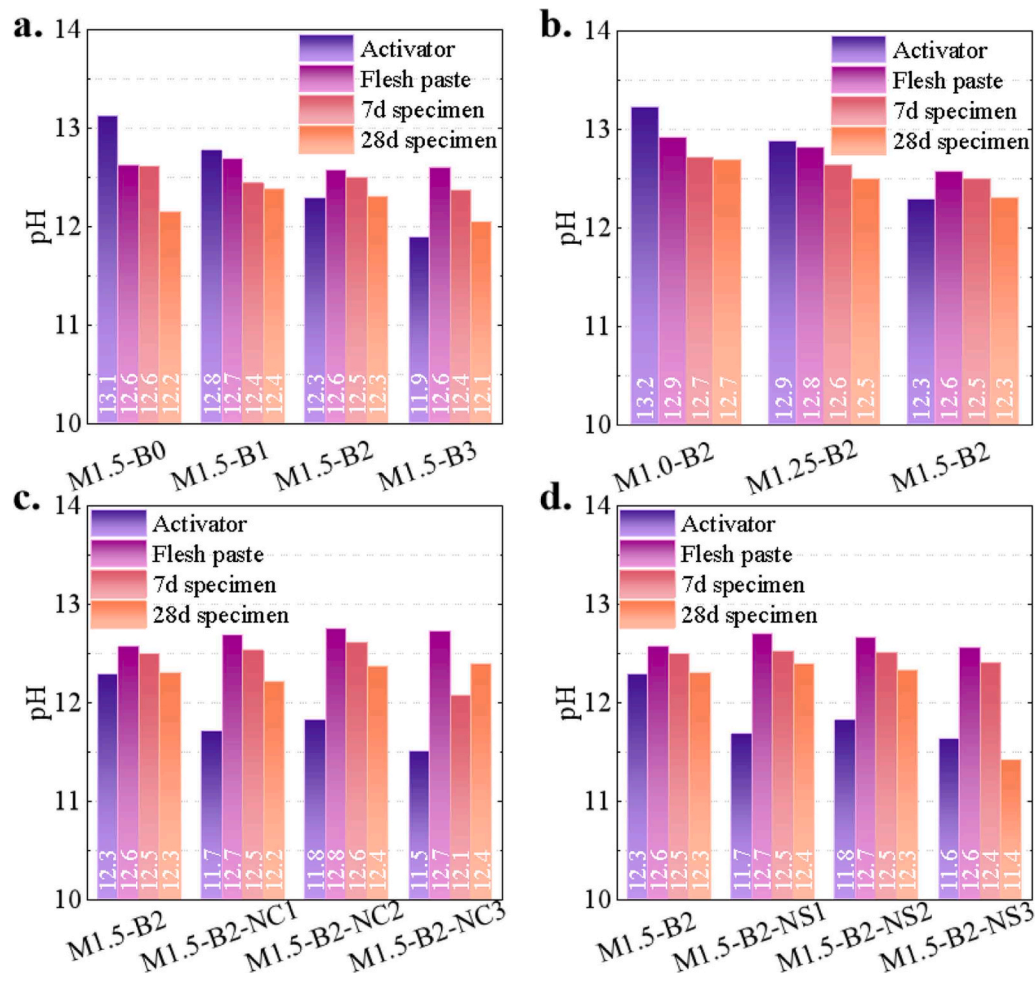


Fig. 9. pH of alkali activator solution, fresh mortar, and hardened blocks of AAM with various activator system parameters: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

In Fig. 10f, a small amount of needle-like ettringite was observed in the AAM containing NS. When excessive NS was used, however, a substantial amount of unreacted CaSO_4 was detected (as confirmed by the EDS elemental analysis in Fig. 10g), indicating insufficient Al to convert CaSO_4 into ettringite. This suggests that the participation of the precursor was limited, which explains the relatively low mechanical performance observed in M1.5-B2-NS3.

Previous studies have indicated a positive correlation between the fracture toughness of AAM and the Ca/Si ratio; therefore, Ca/Si can serve as an important parameter when evaluating the mechanical performance of AAM. In this study, polished AAM specimens were analyzed under backscattered electron (BSE) mode using point-scanning EDS to quantify the Ca/Si ratio in hardened AAM. According to Chen et al. [58], C-A-S-H can be accurately identified at an accelerating voltage of 15 kV with a spatial resolution of 1–2.5 μm . Accordingly, point analyses were conducted at a magnification of $15,000\times$. Points were selected primarily within the C-A-S-H gel regions, avoiding overlap with unreacted slag or fly ash particles. For each specimen, at least 40 valid points were collected across four different regions.

Fig. 11 presents the Ca/Si ratios of AAM under different activator systems using box plots. Fig. 11a indicates that NB incorporation exerted a negligible effect on the Ca/Si ratio, which remained between 0.684 and 0.707. Fig. 11b illustrates the effect of M_s on the Ca/Si ratio. As M_s increased, the Ca/Si ratio gradually decreased from 0.853 to 0.715, which can be attributed not only to the overall reduction of Ca/Si in the system but also to differences in the reaction extent, as confirmed in Section 4.6.2. The addition of NC and NS in the activator system

increased the Ca/Si ratio (Figs. 13c and 13d). However, this did not correspond to an increase in fracture toughness. Microstructural analysis in Fig. 10 suggests that factors beyond Ca/Si, including differences in microstructure, should also be considered when evaluating the fracture toughness of AAM.

4.5.2. Characterization of precursor reaction degree based on SEM

The correspondence between the previously observed elemental distributions and the macroscopic mechanical properties is inconsistent with earlier studies, warranting further investigation. In this work, image processing techniques were employed to quantitatively analyze the degree of precursor reaction in AAM under different activator systems, aiming to elucidate the underlying reasons for the variations in mechanical performance. Fig. 12 illustrates the workflow for characterizing the precursor reaction degree using image processing.

First, grayscale images of AAM were captured in backscattered electron (BSE) mode at a magnification of $5000\times$ to minimize interference from quartz sand. Next, the grayscale distribution curves of the images were deconvoluted using Peak Fit (v4.12) software, and the intersections of the resulting Gaussian curves were defined as the boundaries between different phases. These intersection grayscale values were then used as segmentation thresholds to render the different phases in distinct colors using Photoshop. Finally, the slag phase was extracted, and the area fraction of unreacted slag was quantified using ImageJ to determine the reaction degree of GGBS, α_{GGBS} , as expressed in Eq. (3):

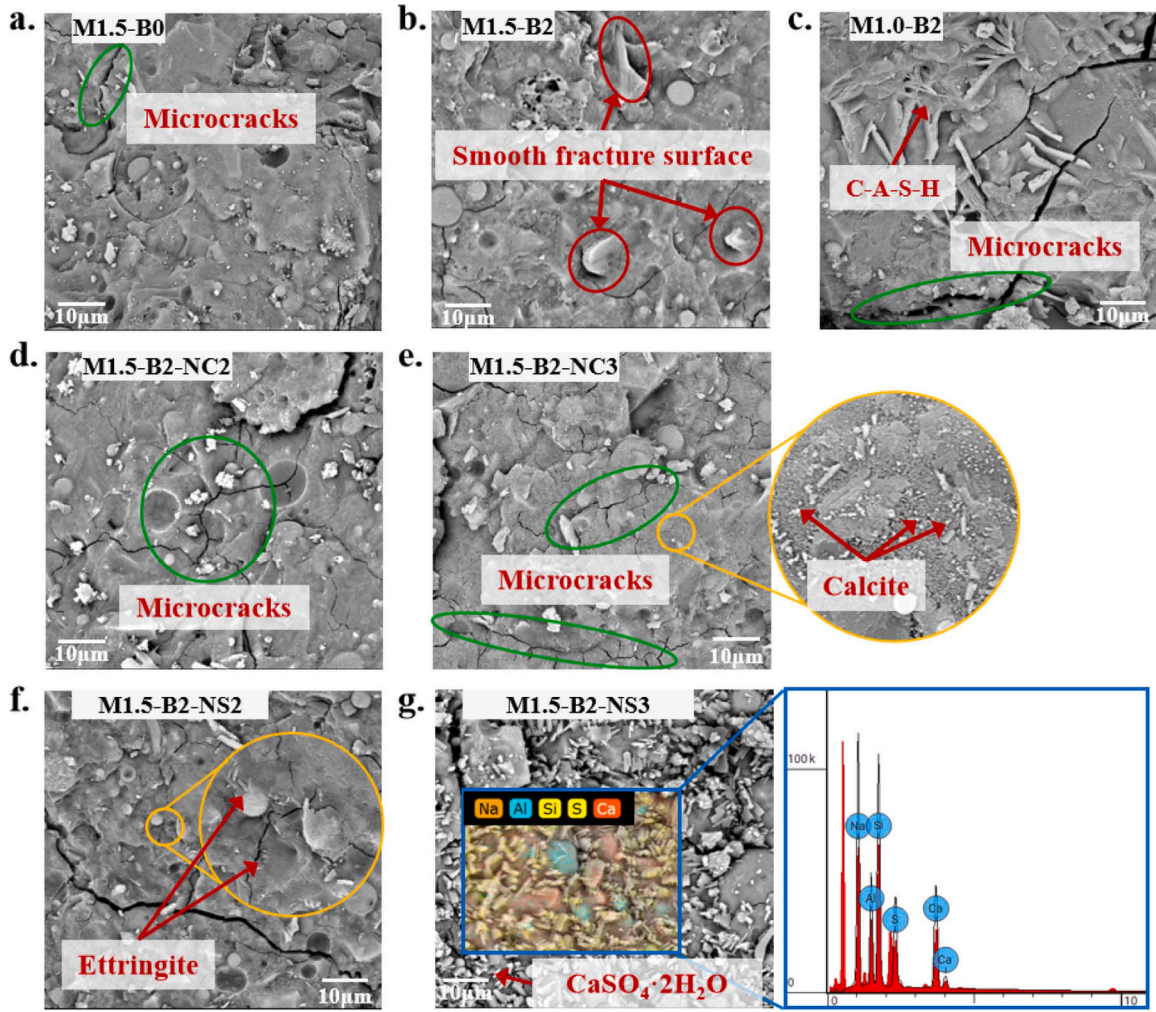


Fig. 10. Microstructure of alkali-activated mortar under different activator systems based on SEM. (a) M1.5-B0; (b) M1.5-B2; (c) M1.0-B2; (d) M1.5-B2-NC2; (e) M1.5-B2-NC3; (f) M1.5-B2-NS2; (g) M1.5-B2-NS3.

$$\alpha_{GGBS} = \left(1 - \frac{V_{L,GGBS}}{V_{0,GGBS}}\right) \times 100\% \quad (3)$$

Here, $V_{L,GGBS}$ is the volume fraction of unreacted slag at age t obtained from BSE image analysis (in this study, only the 28-day specimens were tested), and $V_{0,GGBS}$ is the initial volume fraction of slag in each mix.

However, due to limitations in the polishing procedure and other factors, it is difficult to precisely distinguish between fly ash and the reaction products. Therefore, in this study, the degree of fly ash reaction was indirectly determined by combining the slag reaction degree with the previously obtained Ca/Si ratio, which can be calculated by back-calculating from the Ca/Si formula:

$$R_{Ca/Si} = \frac{(m_{GGBS}\alpha_{GGBS}\omega_{Ca,GGBS} + m_{FA}\alpha_{FA}\omega_{Ca,FA})/n_{Ca}}{(m_{GGBS}\alpha_{GGBS}\omega_{Si,GGBS} + m_{FA}\alpha_{FA}\omega_{Si,FA} + m_{AA}\omega_{Si,AA})/n_{Si}} \quad (4)$$

Here, $R_{Ca/Si}$ denotes the Ca/Si ratio, m_{GGBS} , m_{FA} , and m_{AA} represent the mass fractions of slag, fly ash, and activator in the AAM, respectively. α_{GGBS} and α_{FA} denote the reaction degrees of slag and fly ash, respectively. $\omega_{Ca,GGBS}$ and $\omega_{Ca,FA}$ represent the mass fractions of calcium in slag and fly ash, as obtained from the XRF data in Table 1. $\omega_{Si,GGBS}$, $\omega_{Si,FA}$, and $\omega_{Si,AA}$ denote the mass fractions of silicon in slag, fly ash, and activator, respectively, and n_{Ca} and n_{Si} are the molar masses of calcium and silicon.

Fig. 13 presents the precursor reaction degrees of 28-day AAM under different activator systems. Since the reaction degree of fly ash was

calculated indirectly, its standard deviation is not shown. It can be seen that slag exhibited a much higher reaction degree than fly ash, owing to its substantially higher content of amorphous phase [59], which is corroborated by the XRD results of the raw materials presented in Fig. 1b. Fig. 13a illustrates the effect of NB content on the precursor reaction degree. Both slag and fly ash exhibit an initial increased followed by a decreased in reaction degree with increasing NB content, mirroring the trends observed in compressive strength and fracture toughness. This suggests that the quantity of gel products directly governs the macroscopic mechanical performance of AAM and accounts for the significant enhancement in fracture toughness at 0.5 Na₂O% NB, despite negligible changes in the Ca/Si ratio. Conversely, higher NB contents could reduce alkalinity and suppress precursor reactions. Notably, although the precursor reaction degree in M1.5-B2 was slightly lower than in M1.5-B0, its compressive strength and fracture toughness were higher, suggesting that gels formed through NB participation may possess superior mechanical properties.

Fig. 13b illustrates the effect of M_s on precursor reaction degree. Although M1.25-B2 exhibited the highest reaction degree, it did not correspondingly show the best mechanical performance in terms of compressive strength and fracture toughness. This indicates that in sodium silicate-activated systems, an appropriate Ca/Si ratio has a stronger influence on mechanical properties, with optimal performance observed at a Ca/Si of approximately 0.707. Additionally, the reaction degree of fly ash increased with M_s , consistent with previous findings by Biggeli et al. [60], likely related to the soluble Si content in the system

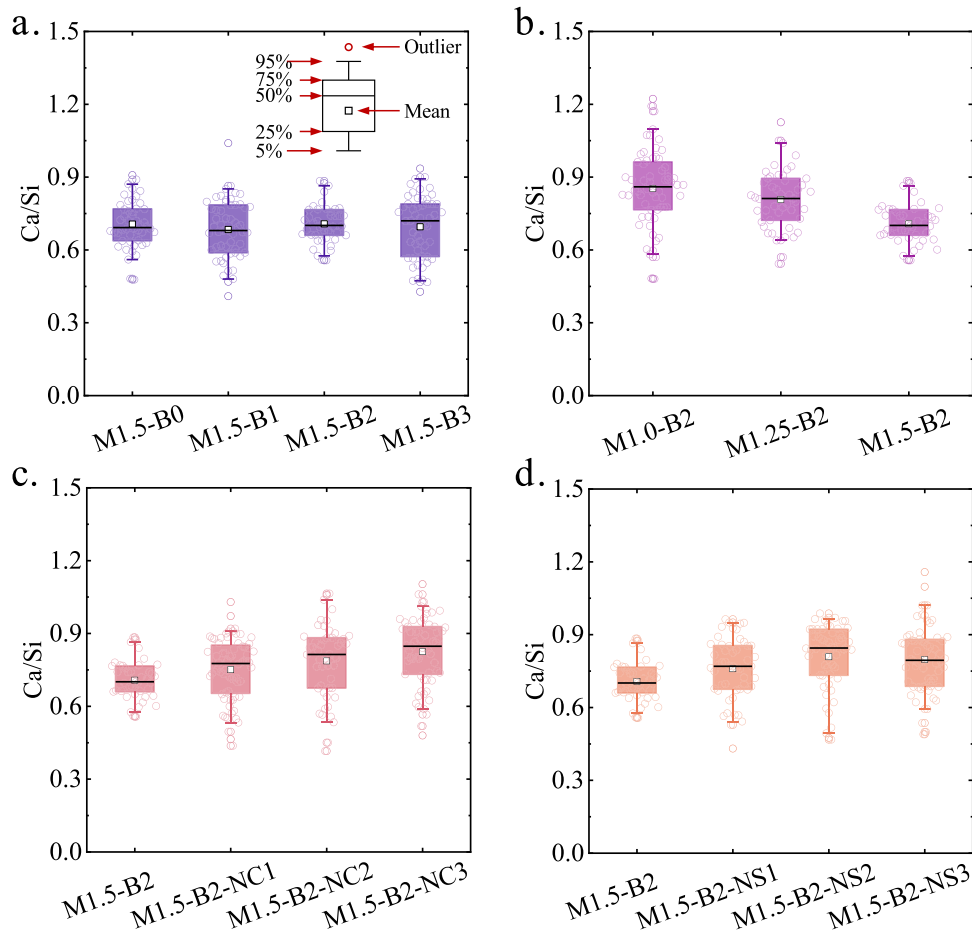


Fig. 11. Box plot of calcium-to-silicon ratio of AAM with various activator system parameters: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

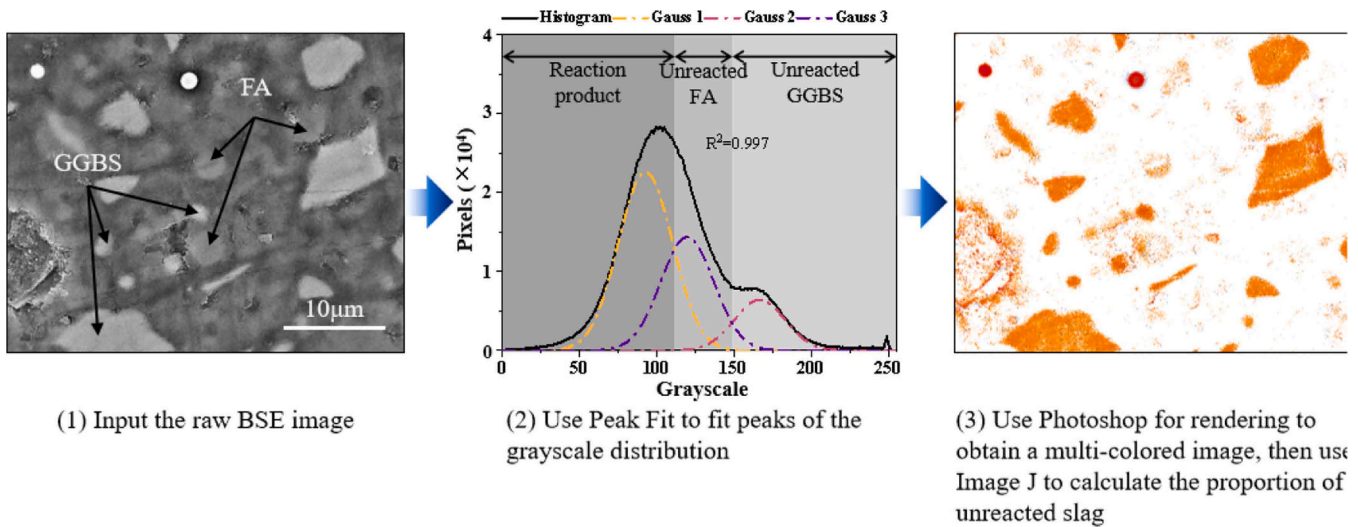


Fig. 12. Process of characterizing the reaction degree of AAM precursors using image processing technology.

[58,59]. The incorporation of NC had little effect on slag reaction degree, but the reaction degree of fly ash decreased with increasing NC content, as shown in Fig. 13c. This reflects the competitive adsorption between CO_3^{2-} and SiO_4^{2-} on Ca^{2+} : the continuously released Ca^{2+} during hydration is preferentially captured by anions to form precipitates, thereby promoting slag reaction. However, the pH reduction caused by higher NC content hinders the dissolution of the inherently less reactive fly ash. Consequently, in NC-containing AAM, the reduction in gel

products and the increase in microcracks within the gel structure appear to be the main factors contributing to the observed decrease in fracture toughness.

Fig. 13d illustrates the effect of NS on precursor reaction degree. As the NS content increased, the reaction degrees of both slag and fly ash decreased noticeably. Although M1.5-B2-NS1 exhibited a slightly lower precursor reaction degree compared to M1.5-B2. Its improved macroscopic mechanical performance is likely attributable to the formation of

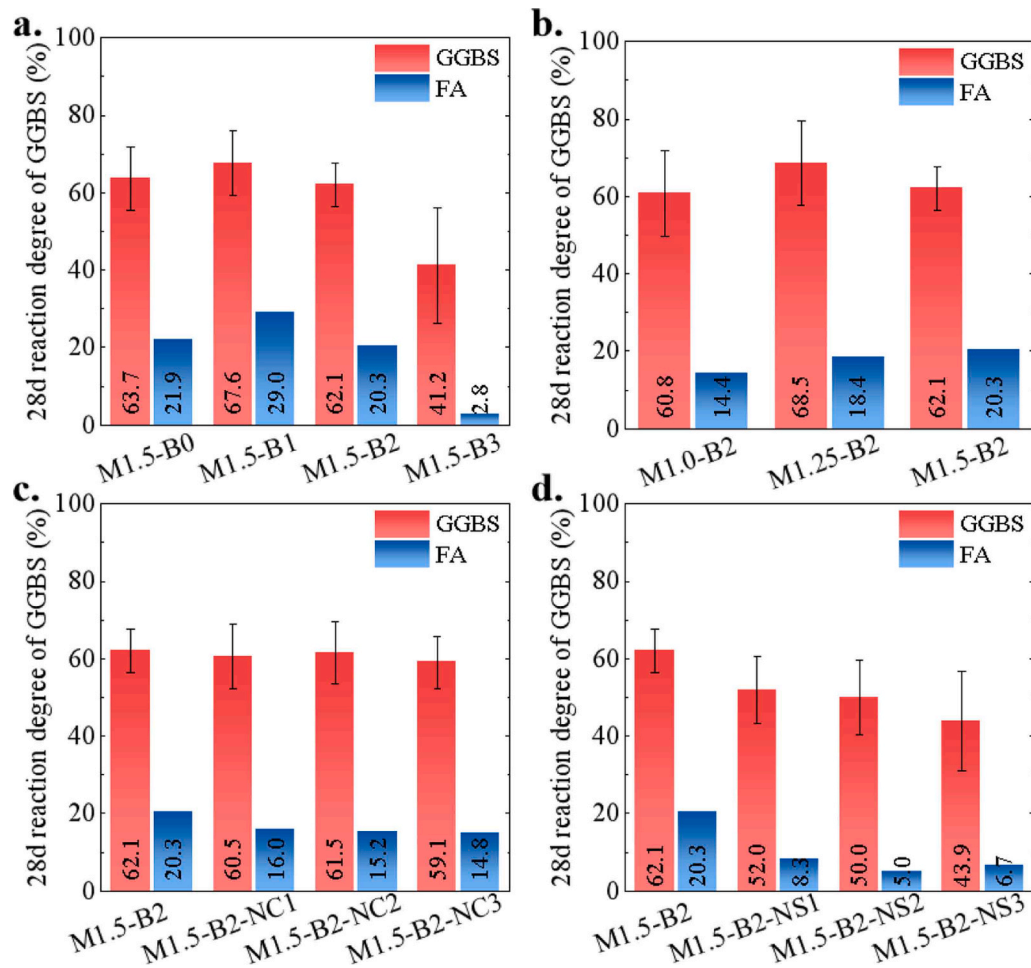


Fig. 13. Reaction degree of AAM precursors at 28d age with various activator system parameters: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

a small amount of ettringite, which enhances the compactness of the AAM binder structure. However, at higher NS contents, both precursor reaction degree and macroscopic mechanical properties are significantly reduced. This can be attributed to the lower activation efficiency of NS compared to sodium silicate. At high NS contents, the material cannot fully participate in the hydration reaction, as evidenced by the microstructure in Fig. 10g. Consequently, gel production is reduced and leading to insufficient bonding with aggregates and a decline in mechanical performance. Therefore, the optimal content range of NS is considerably narrower than that of NC, and it should only be used in small amounts.

4.5.3. Mineralogical characterization

XRD was employed to characterize the mineralogical composition of AAM under different activator systems, as shown in Fig. 14. Across all activator systems, distinct diffraction peaks were observed in the $26\text{--}28^\circ$ 2θ range, primarily corresponding to unreacted mullite (PDF# 01-089-2644) and quartz (PDF# 01-074-1811) in the fly ash. In addition, a notable diffraction peak around 30° 2θ was detected, which is attributed to the main gel product of AAM, C-(A)-S-H (PDF# 00-019-0052 or PDF# 00-029-0331) [26,61,62]. The broad nature of this peak indicates poor crystallinity, suggesting that C-(A)-S-H predominantly exists in an amorphous form.

Figs. 16a and 16b demonstrate that the crystalline phases remain largely unaffected by NB content or M_s , with no new gel products detected, in agreement with previous studies [63,64]. However, in Fig. 14b, the diffraction peak of C-(A)-S-H became sharper as M_s increased, suggesting a slight enhancement in the crystallinity of the gel product. Fig. 14c shows that the incorporation of NC led to the formation

of a new calcite phase (PDF# 01-072-1650), whose main peak overlaps with that of C-(A)-S-H [26,65]. This is corroborated by the increasing intensity of the diffraction peak at 29.6° 2θ with higher NC content and NC's negative effect on the degree of hydration. In Fig. 14d, the appearance of ettringite is observed near 22.9° 2θ (PDF# 00-041-1451). However, the peak is relatively weak due to the low content or crystallinity of the product. Notably, the diffraction peak of C-(A)-S-H gradually sharpened with increasing NS content, indicating that NS may promote Al^{3+} dissolution and enhance the crystallinity of C-(A)-S-H.

4.5.4. Pore structure analysis

The pore structure of selected 28-day AAM specimens was analyzed using MIP, with the results presented in Fig. 15. Fig. 15a shows that the cumulative pore volume curve of mix M1.5-B0 is positioned highest, with a steeper slope observed in the larger pore diameter range. This indicates a higher total pore volume and a greater proportion of large-diameter pores, primarily attributable to its poor workability. In contrast, the pore size distributions of the other mixes are relatively similar. Pores of different sizes have distinct effects on the mechanical properties of concrete or mortar. Based on diameter, pores can be categorized as follows: harmless pores ($d < 20$ nm), less harmful pores (20 nm $< d < 50$ nm), harmful pores (50 nm $< d < 200$ nm), and more harmful pores ($d > 200$ nm) [66].

The AAM samples tested in this study exhibited pore size distributions across all ranges, with relatively high proportions of pores both below 10 nm and above 200 nm (as shown in Fig. 15b). Mix M1.5-B0 showed the highest porosity, reaching 20.5%. After the addition of NB, the fluidity of the paste improved significantly, and the porosity

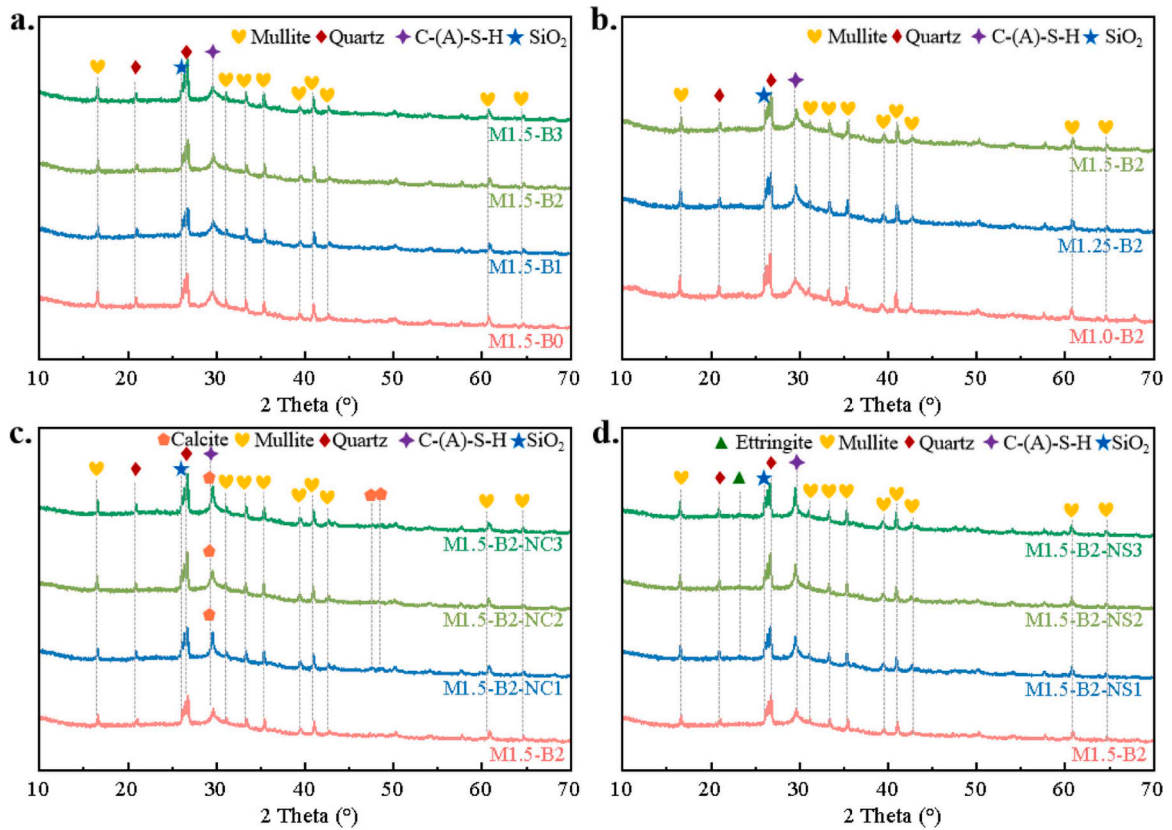


Fig. 14. XRD patterns of AAM at 28d age with various activator system parameters: (a) NB content, (b) M_s , (c) NC content, and (d) NS content.

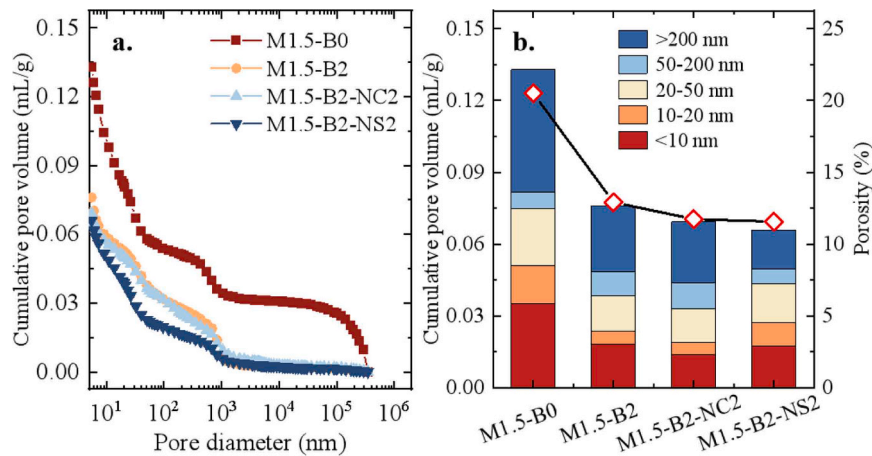


Fig. 15. Pore structure of AAM at 28-day age under different activator systems. a. Cumulative pore volume curves; b. Pore size distribution and porosity.

decreased to between 11.6 % and 12.9 %. The dense pore structure and low porosity contributed to the high compressive strength of the AAM.

5. Correlation analysis of AAM performance

To further investigate the relationships between variables and performance parameters in AAM research, Spearman correlation analysis was employed to examine the correlations among activator parameters (NB content, M_s , NC content, NS content), reaction progress parameters (α_{GGBS} , α_{FA}), fresh-state properties ($ST_{initial}$, ST_{final} , Flowability), and macroscopic mechanical properties (f_c , G_F , K_m). Spearman correlation (ρ) is commonly used to assess whether a variable tends to increase or decrease in tandem with another [67]. A correlation coefficient with an absolute value greater than 0.50 indicates a significant monotonic

relationship between the variable and the output [68].

Fig. 16 presents a heatmap that clearly illustrates both the strength and direction of correlations among the variables. The analysis reveals several highly significant correlations. Regarding reaction progress, α_{GGBS} and α_{FA} exhibit very strong negative correlations with NB content ($\rho = -0.81$ and -0.78 , respectively), suggesting that excessive NB content may inhibit the dissolution and reaction of precursors. In contrast, correlations involving other activator parameters are relatively weak. Moreover, α_{GGBS} and α_{FA} are strongly positively correlated ($\rho = 0.92$), indicating a synergistic rather than independent alkali-activation reaction between GGBS and FA in this experimental system.

Setting time is significantly influenced by the activator composition. NB, NC, and NS contents show strong positive correlations with setting time ($\rho > 0.7$), implying that increasing the dosage of these activators can

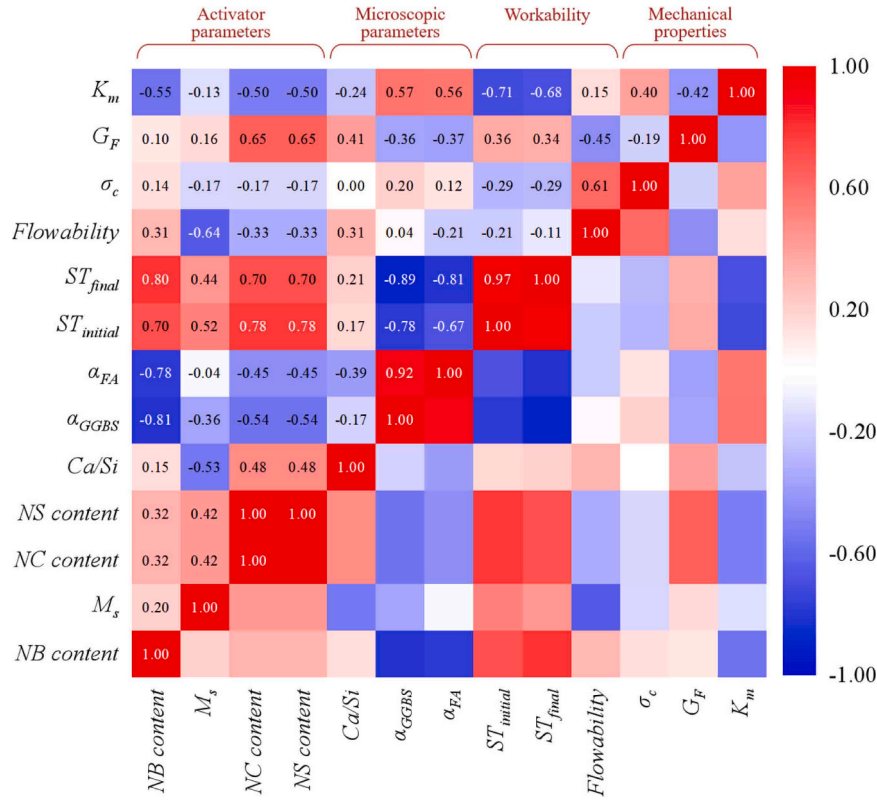


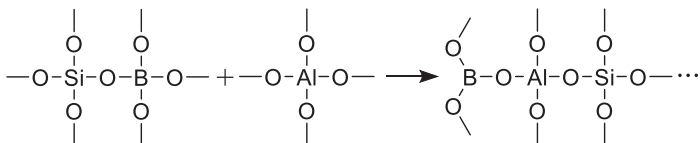
Fig. 16. Correlation matrix of workability, and mechanical performance.

substantially prolong the setting process.

Regarding mechanical performance, compressive strength is positively correlated with flowability ($\rho=0.61$), likely because good flowability ensures adequate compaction during material molding. Fracture energy is positively correlated with NC and NS contents ($\rho=0.65$), consistent with MIP observations indicating their role in reducing porosity. Conversely, fracture toughness shows a negative correlation with NC and NS contents. However, no significant direct correlations were observed between macroscopic mechanical properties and microscopic reaction degrees, suggesting that the performance of AAMs is governed by the complex interplay of microstructural evolution, types and distributions of reaction products, and other factors, rather than being solely determined by a single reaction degree parameter.

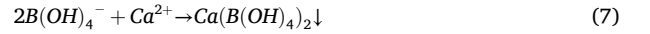
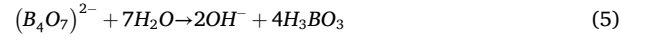
6. Mechanistic analysis of activator systems on AAM performance

(1) Effect of NB the performance of AAM



(8)

As shown in Fig. 17a, the incorporation of NB into fresh AAM could lead to the formation of calcium borate [42,58,69], and it would coat the surface of the slag particles. This process can be represented by Eqs. (5), (6), and (7):



Here, the calcium borate acts as a semi-permeable gel layer, allowing only water molecules and small ions to pass while hindering the release of Ca, Si, and Al. This inhibits the hydration of the precursors and prolongs the setting time. Meanwhile, the calcium borate layer imparts a negative zeta potential to the AAM, enhancing its workability. As hydration progresses, the calcium borate film gradually breaks down [43], releasing Ca^{2+} and aluminosilicate species. The reaction continues, with some Al-O bonds being replaced by B-O bonds, forming borosilicate-aluminate gel products [49]:

The formation of B-O bonds positively affects the mechanical performance of AAM, manifested as increases in compressive strength and fracture toughness. However, excessive NB addition significantly lowers the pore solution pH, inhibiting precursor

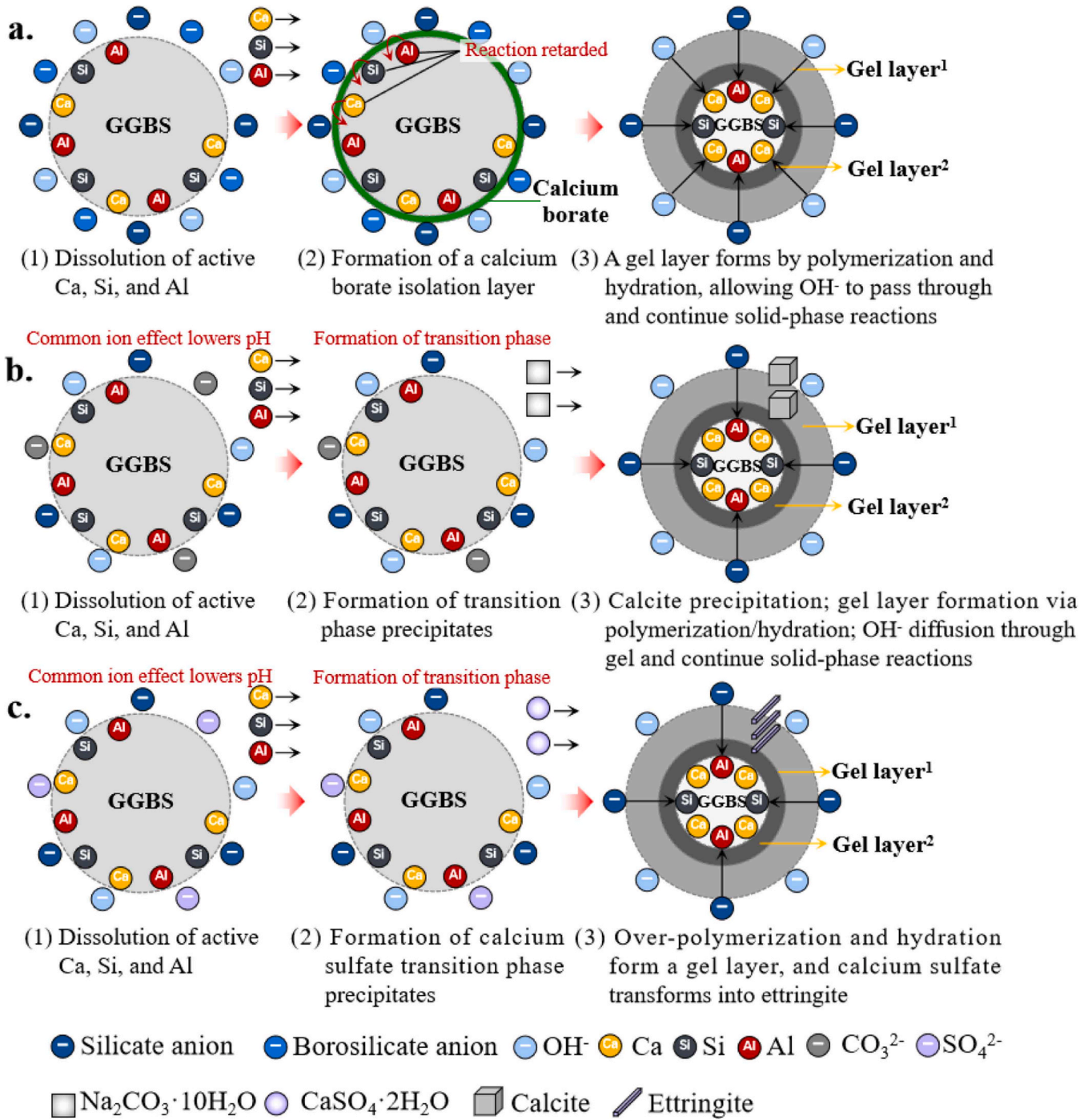


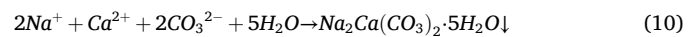
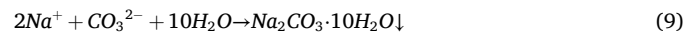
Fig. 17. Reaction process and products of AAM under different activator systems. (a) Effect of borax; (b) Effect of Na₂CO₃; (c) Effect of Na₂SO₄.

hydration and reducing the mechanical properties of AAM. XRD and SEM observations indicate that NB incorporation enhances the crystallinity of the gel products, and the fracture surfaces exhibit cleavage behavior [49,57], leading to a decrease in fracture energy. In addition, the reduced hydration degree decreases the amount of gel products, which negatively impacts the volumetric and mass stability of the AAM.

(2) Effect of NC on the performance of AAM

As shown in Fig. 17b, under highly alkaline conditions, the solubility of NC is reduced due to the common-ion effect, leading to the formation of Na₂CO₃·10H₂O crystalline precipitates. Meanwhile, a portion of NC reacts with Ca²⁺ in the pore solution to form Na₂Ca(CO₃)₂·5H₂O as a

transient precipitate. The formation of these intermediate phases is illustrated in Eqs. (9) and (10) [3,48]:



Due to the common ion effect, the pH of the sodium carbonate-sodium silicate composite activator solution was lower than that of the control group, which might also be one of the reasons for the delayed coagulation of AAM. However, with the formation of transition phases in the fresh AAM paste (as shown in Eqs. (9) and (10)), Na⁺ and CO₃²⁻ ions were largely consumed. This consumption promoted the dissolution of solid sodium silicate and generated a significant amount of OH⁻,

Table 3

Carbon emission factor and price of raw materials.

Materials	Carbon emission factor, <i>C</i> (kg CO ₂ -e/kg)	Price, <i>P</i> (CNY/kg)
Cement	0.820–0.912 [72–77]	0.700
GGBS	0.019–0.089 [4,78–81]	0.490
FA	0.004–0.200 [4,78–80,82]	0.539
Na ₂ O-SiO ₂	0.705–1.860 [83–86]	5.200
Na ₂ O-2SiO ₂	0.705–1.860 [83–86]	4.300
Na ₂ CO ₃	0.111–0.700 [3,83,86,87]	4.500
Na ₂ SO ₄	0.000–0.322 [73,88]	2.200
Borax	1.310–2.048 [89,90]	3.400
Quartz sand	0.010–0.025 [4,72,91]	0.392
Water	0.001 [79,86,92]	0.002
Super-plasticizer	0.720–1.667 [74–77]	19.00

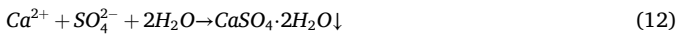
leading to an increase in the pH of the free water:



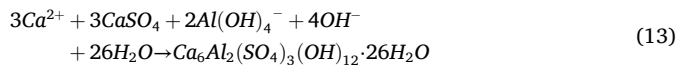
The formation of these transitional precipitates not only reduces the amount of free water but also affects particle packing, thereby decreasing the workability of AAM [13,70]. Additionally, the consumption of Ca²⁺ during the formation of these phases may hinder the early development of the gel network, extending the setting time. As the reaction progresses, the transitional phases gradually dissolve, and the remaining CO₃²⁻ participates in forming calcite, which fills gel pores and densifies the AAM microstructure, as evidenced by the significant reduction in porosity observed via MIP. Consequently, the dense microstructure resulting from an appropriate NC content enhances both the compressive strength and fracture energy of AAM. However, the formation of relatively low-density calcite also introduces additional microcracks in the gel matrix, leading to a reduction in fracture toughness.

(3) Effect of NS on the performance of AAM

As shown in Fig. 17c, the incorporation of NS leads to the formation of calcium sulfate transitional precipitates. These precipitates later serve as the reactants for ettringite formation:



Similarly, the formation of calcium sulfate reduces the amount of free water, lowers workability, and delays setting by hindering the dissolution and polycondensation of aluminosilicates [45]. As hydration proceeds, free Ca²⁺ and aluminates in the pore solution react with calcium sulfate to form ettringite:



The formation of a small amount of ettringite provides additional

pore filling, and this in turn enhances the compressive strength, fracture toughness, and fracture energy of M1.5-B2-NS1 at low NS content. In contrast, excessive NS generates large quantities of unreacted CaSO₄·2H₂O, severely hindering the development of mechanical properties.

7. Comprehensive analysis of sustainability and mechanical performance

Sodium silicate, NB, NC, and NS serve as activators that not only influence the mechanical and fresh-state properties of alkali-activated binders differently but also affect the economic and environmental performance of AAM due to their varying costs and carbon footprints. In this section, the carbon emissions and costs of AAM are calculated separately. Compressive strength is used as a performance metric to evaluate the combined economic and environmental benefits via a comprehensive carbon emission index (*CI*, defined as the ratio of carbon emission to strength) and a comprehensive cost index (*CCI*, defined as the ratio of cost to strength). A conventional cement-based mortar [71] was selected for a sustainability comparison. Carbon emissions and costs are calculated using Eqs. (14) and (15), while the comprehensive indices are determined using Eqs. (16) and (17):

$$CE = \sum_i^n C_i M_i \quad (14)$$

$$\text{Cost} = \sum_i^n P_i M_i \quad (15)$$

$$CI = CE / \sigma_c \quad (16)$$

$$CCI = \text{Cost} / \sigma_c \quad (17)$$

Here, *CE* and *Cost* represent the carbon emissions (kg CO₂-e/m³) and cost (CNY/m³) of the AAM, respectively; *C_i* is the carbon emission factor of raw material *i* (kg CO₂-e/kg); *M_i* is the mass of raw material *i* per cubic meter of AAM (kg/m³); *P_i* is the price of raw material *i* (CNY/kg); and *σ_c* is the 28-day compressive strength. Table 3 lists the carbon emission factors and prices of the raw materials, while Table 4 presents the calculated *CI* and *CCI* values.

Fig. 18 presents the comprehensive performance indicators of AAM under different activator systems. As shown in Figs. 18a and 18b, All AAMs proposed in this study demonstrate pronounced advantages in carbon emission reduction compared to the cement-based mortar, while they lag behind in cost-effectiveness; NB has a limited effect on reducing carbon emissions but significantly lowers costs; however, excessive addition (1.5 Na₂O %) negatively impacts both cost and emissions. For one-step alkali-activated mortars, reducing *M_s* decreases activator usage, favoring both carbon reduction and cost savings. For NC and NS, a dosage of 0.375 Na₂O % yields the lowest cost and carbon emissions while achieving optimal compressive strength. Therefore, partially replacing sodium silicate with a small amount of NC or NS is an effective

Table 4

Comprehensive performance indicators of AAM.

Mix IDs	<i>CE_{min}</i>	<i>CE_{max}</i>	<i>Cost</i>	<i>CI_{min}</i>	<i>CI_{max}</i>	<i>CCI</i>	<i>MEI</i>
Hou et al. (M1)	599	817	1161	8.09	11.0	15.7	321
M1.5-B0	109	404	1304	1.98	7.38	23.8	551
M1.5-B1	150	469	1413	2.42	7.55	22.8	427
M1.5-B2	192	534	1523	2.82	7.85	22.4	529
M1.5-B3	233	599	1633	3.80	9.75	26.6	1747
M1.0-B2	173	484	1460	2.98	8.35	21.5	743
M1.25-B2	182	509	1495	2.73	7.62	22.4	769
M1.5-B2-NC1	185	520	1516	2.44	6.84	20.0	1070
M1.5-B2-NC2	179	506	1502	2.50	7.05	20.9	1110
M1.5-B2-NC3	167	479	1471	3.30	9.46	29.1	855
M1.5-B2-NS1	185	518	1511	2.45	6.86	20.0	498
M1.5-B2-NS2	178	503	1487	2.67	7.56	22.4	872
M1.5-B2-NS3	164	472	1436	3.47	9.97	30.3	929

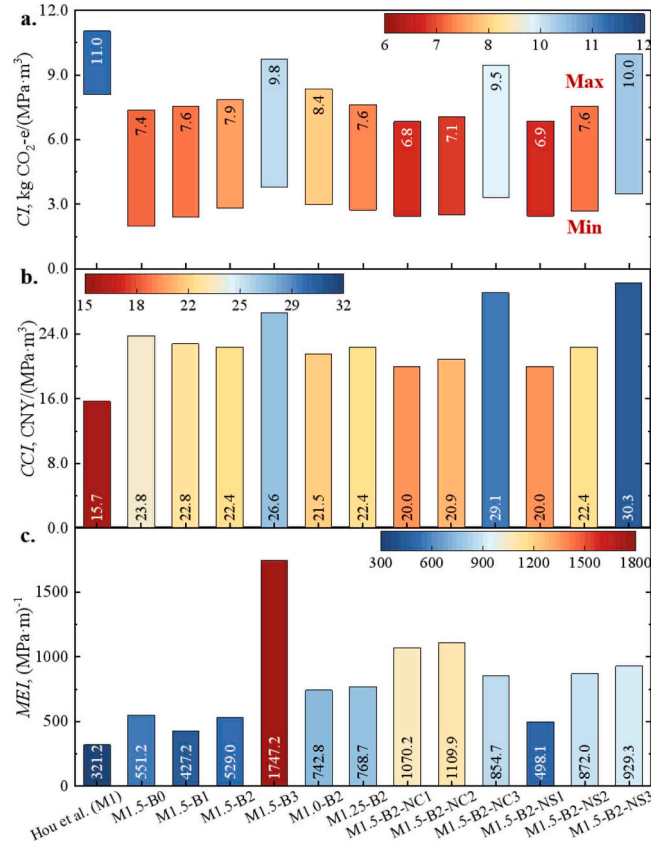


Fig. 18. Comprehensive performance indicators of AAM: (a) carbon emissions; (b) cost; (c) matrix evaluation indicator.

approach to producing low-carbon, high-performance alkali-activated materials.

To further explore the suitable application scenarios for different composite activators, Engineered geopolymer composites is taken as an example for analysis. According to the microstructural design criteria of Engineered geopolymer composites [30–32], to ensure stable crack propagation under tensile loading, the fiber bridging energy J'_b must exceed the crack-tip fracture toughness J_{tip} . The latter can be approximated using the matrix fracture toughness K_m and elastic modulus E_m :

$$J_{tip} = \frac{K_m^2}{E_m} \quad (18)$$

J_b can be indirectly enhanced by improving the matrix strength and density [32,93]. Therefore, this study proposes using the ratio of compressive strength σ_c to the square of the matrix fracture toughness as a matrix evaluation index (MEI) to assess the potential of a matrix for Engineered geopolymer composites development. A higher MEI indicates a greater potential for stable crack propagation in EGC. The specific calculation results are presented in Table 4.

$$MEI = \frac{\sigma_c}{K_m^2} \quad (19)$$

As can be seen from Fig. 18c, all AAMs proposed in this study exhibit higher MEI values than the cement-based mortar, demonstrating their potential for producing high-ductility concrete. The highest MEI among the AAMs was achieved by M1.5-B3, which is attributed to the weakening effect of excessive NB on fracture toughness. AAM incorporating small amounts of NC or NS also achieves relatively high MEI values. Considering the combined cost and carbon emission metrics, the addition of NC as an activator emerges as a reliable strategy to enhance both the sustainability and mechanical performance of Engineered geopolymer composites. It should be noted that the sustainability analysis in this study is a preliminary assessment, focusing primarily on material-

level carbon emissions and costs. A comprehensive life cycle assessment (LCA) following ISO standards and encompassing the entire life cycle is required to obtain a more complete sustainability profile. Accordingly, conducting in-depth and systematic LCA research represents an important direction for future work.

8. Conclusion

This study systematically investigated the workability and mechanical performance of slag/fly ash-based AAM prepared with various solid activator combinations (anhydrous sodium silicate, NB, sodium carbonate, and sodium sulfate), and explored the mechanisms by which composite activators influence the microstructure of AAM. Based on the results, the main conclusions are summarized as follows:

- At a NB content of 1.0 Na₂O%, AAM exhibited a balanced combination of good workability, appropriate setting time, and high mechanical performance. NB forms a calcium-borate protective layer on the slag surface, effectively delaying early hydration and improving workability. At an optimal content, this retardation promotes uniform formation of reaction products, ensuring strength development. However, excessive NB significantly lowers the system pH, hinders precursor hydration, and reduces gel formation, resulting in decreased compressive strength and fracture performance.
- At a sodium silicate modulus of 1.5, the AAM system would show moderate alkalinity, complete precursor reaction, abundant gel formation, and a dense microstructure, resulting in a well-balanced combination of setting time, compressive strength, and fracture toughness. Reducing the modulus to 1.0–1.25 could still yield high mechanical performance, but the faster reaction rate increased the risk of flash setting, limiting practical applicability.
- Low additions (0.375 Na₂O%) of sodium carbonate or sodium sulfate promote the formation of a dense and uniform gel structure and

optimize pore distribution, thereby enhancing AAM strength. However, excessive addition reduces system alkalinity, hinders the transformation of intermediate phases, and causes incomplete reactions, leading to decreased mechanical performance.

- The combined evaluation of mechanical performance and sustainability indicates that both sodium carbonate and sodium sulfate hold potential for developing low-carbon, high-performance construction materials. AAM activated with a sodium carbonate-sodium silicate blend exhibits a “high-strength, low-toughness” profile, making it suitable for engineering geopolymer composites. In contrast, the sodium sulfate-sodium silicate system shows consistent improvements in both compressive strength and fracture toughness, making it more appropriate for limited incorporation into ordinary alkali-activated concrete to enhance overall mechanical performance.

CRediT authorship contribution statement

Yong-Chang Guo: Writing – review & editing, Project administration, Methodology. **Jia-Xiang Lin:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis. **Yong-jian Cai:** Writing – original draft, Software, Methodology, Investigation. **Rui-hao Luo:** Writing – original draft, Visualization, Validation. **Gai Chen:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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